



Elixirr



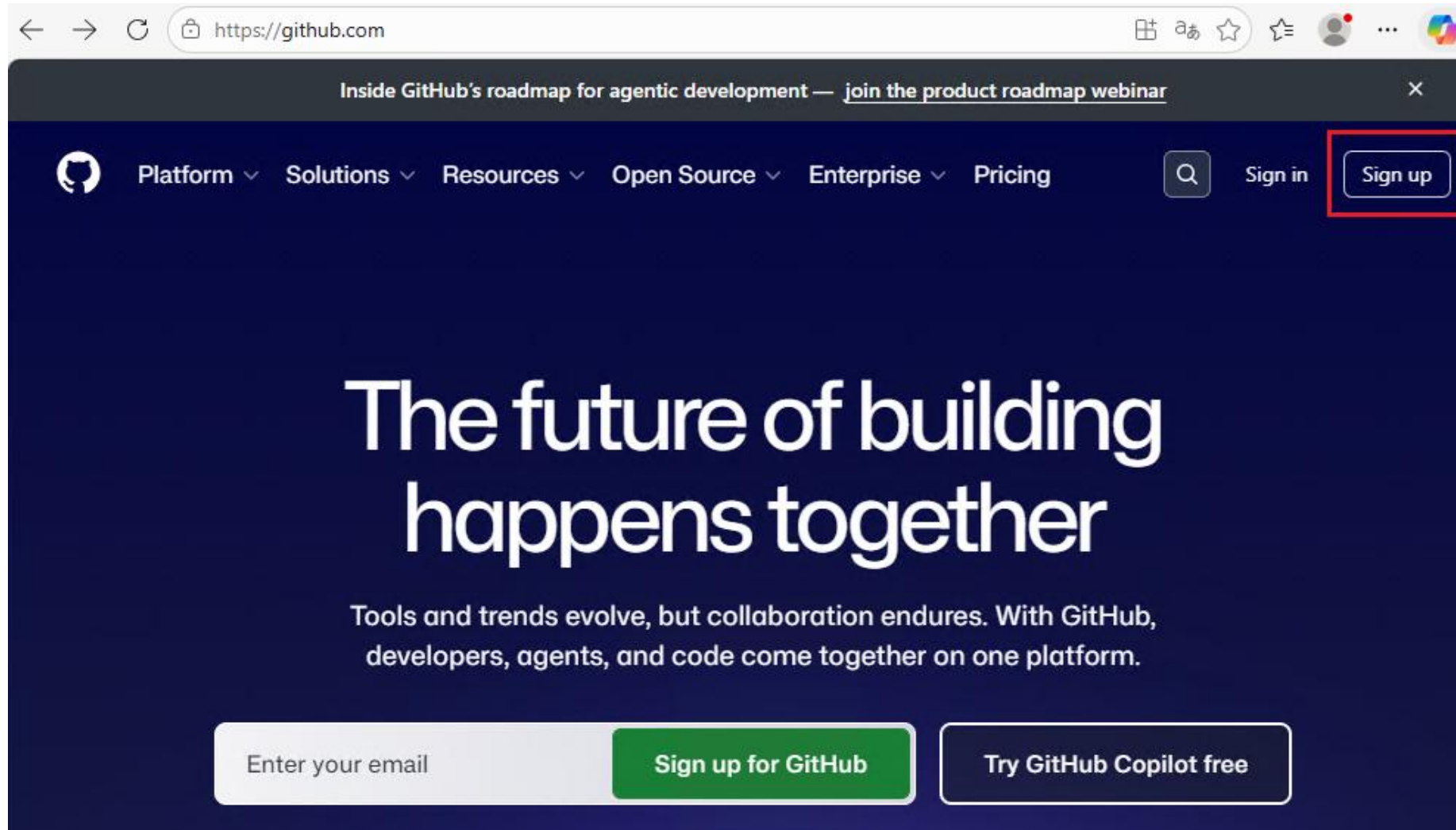
개발 환경 설정 5

파이썬 가상 환경

본 자료의 일부 또는 전체는 Elixirr의 사전 서면 승인 없이 외부로 배포, 인용 또는 복제하는 행위를 금합니다.

GitHub Sign up 1

<https://github.com/>



GitHub Sign up 2

Create your free account

Explore GitHub's core features for individuals and organizations.

[See what's included](#) ▾

Already have an account? [Sign in](#) →

Sign up for GitHub

 Continue with Google

 Continue with Apple

or

Email*

Password*

Password should be at least 15 characters OR at least 8 characters including a number and a lowercase letter.

Username*

Username may only contain alphanumeric characters or single hyphens, and cannot begin or end with a hyphen.

Your Country/Region*

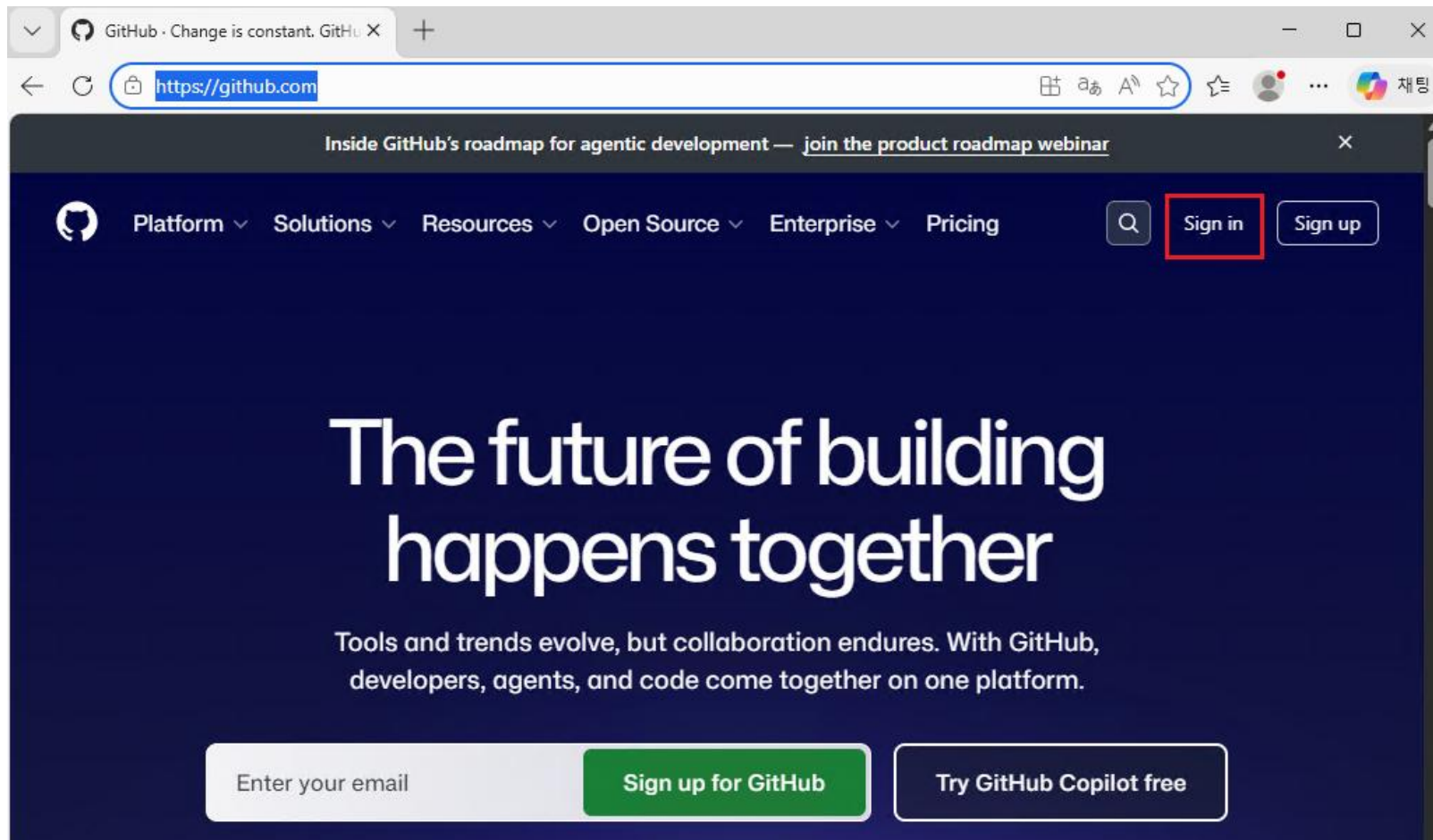
Korea, South ▾

For compliance reasons, we're required to collect country information to send you occasional updates and announcements.

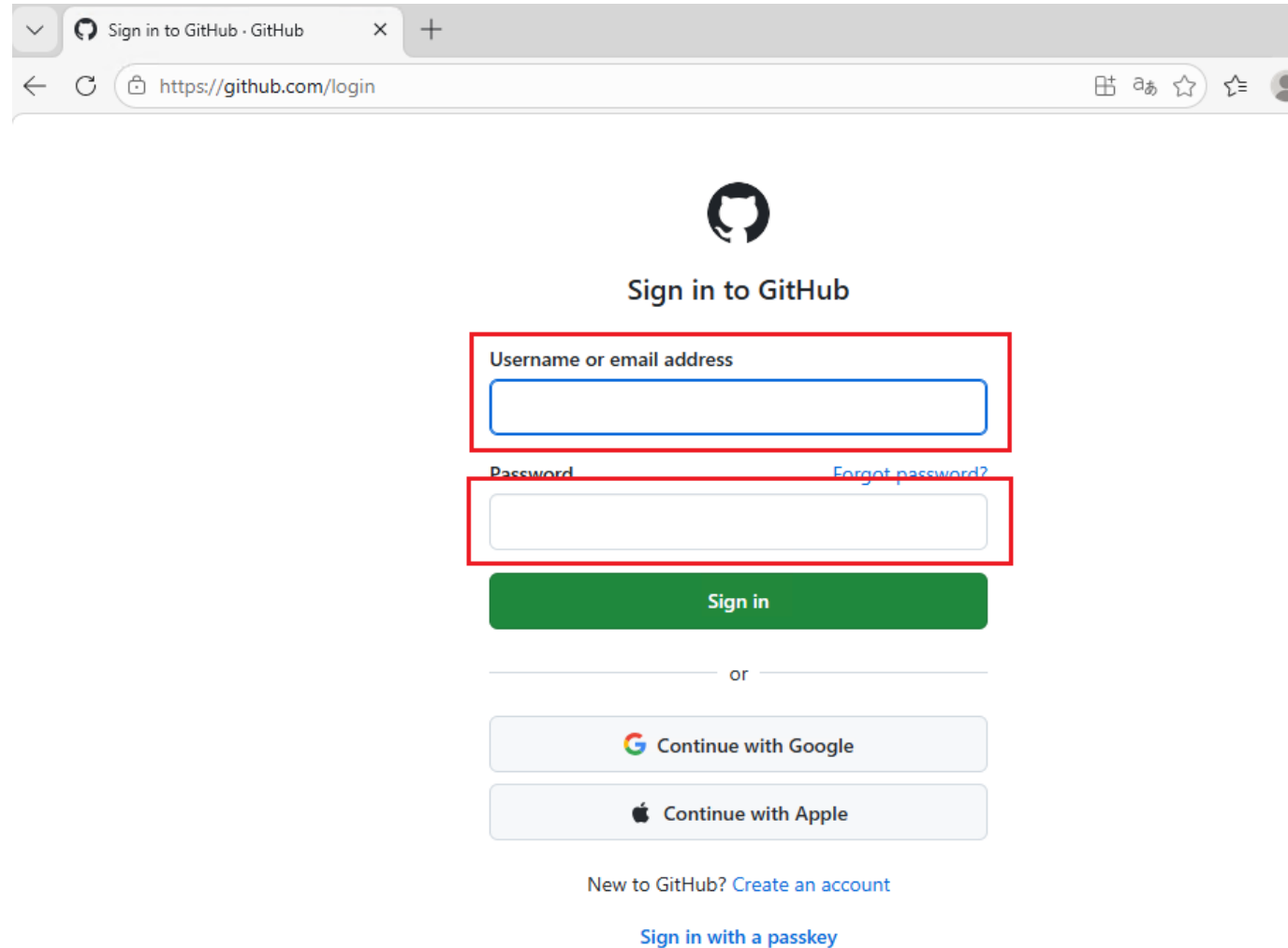
Create account >

GitHub Sign in 1

<https://github.com/>



GitHub Sign in 2



The screenshot shows the GitHub login page in a web browser. The browser's address bar displays "https://github.com/login". The page features the GitHub logo at the top, followed by the heading "Sign in to GitHub". Below this, there are two input fields: "Username or email address" and "Password". Both fields are highlighted with red rectangular boxes. To the right of the password field is a link that says "Forgot password?". Below the input fields is a green "Sign in" button. Underneath the button is a horizontal line with the word "or" in the center. Below this line are two buttons: "Continue with Google" and "Continue with Apple". At the bottom of the page, there is a link that says "New to GitHub? Create an account" and another link that says "Sign in with a passkey".

Sign in to GitHub · GitHub

https://github.com/login

Sign in to GitHub

Username or email address

Password [Forgot password?](#)

Sign in

or

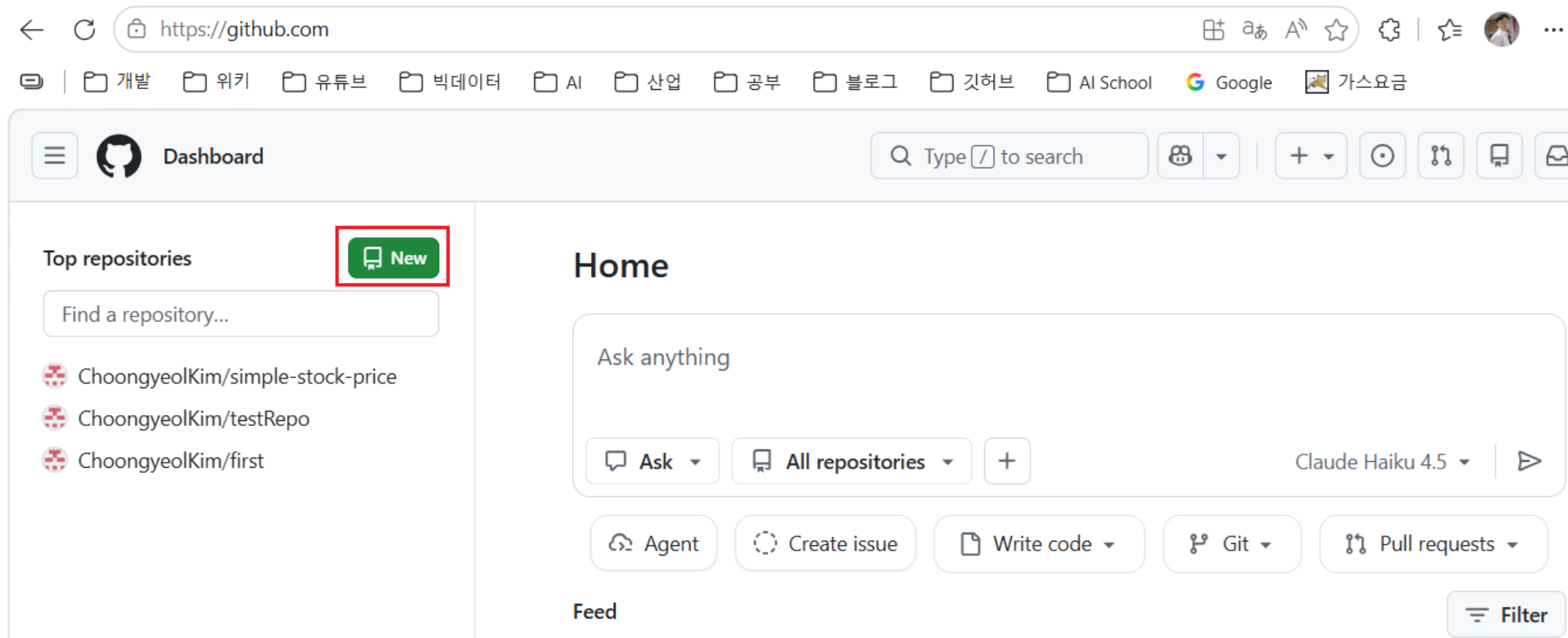
 Continue with Google

 Continue with Apple

New to GitHub? [Create an account](#)

[Sign in with a passkey](#)

프로젝트 코드 원격 저장소 만들기 1



The screenshot shows the GitHub Dashboard interface. At the top, there's a navigation bar with the GitHub logo, the word 'Dashboard', a search bar, and several utility icons. Below this, a row of bookmarks includes '개발', '위키', '유튜브', '빅데이터', 'AI', '산업', '공부', '블로그', '깃허브', 'AI School', 'Google', and '가스요금'. The main content area is divided into two columns. The left column, titled 'Top repositories', contains a search bar 'Find a repository...' and a list of repositories: 'ChoongyeolKim/simple-stock-price', 'ChoongyeolKim/testRepo', and 'ChoongyeolKim/first'. A green 'New' button with a repository icon is highlighted with a red box. The right column, titled 'Home', features a large 'Ask anything' section with a search bar, a dropdown menu set to 'All repositories', and a 'Claude Haiku 4.5' model selector. Below this are buttons for 'Agent', 'Create issue', 'Write code', 'Git', and 'Pull requests'. At the bottom, there's a 'Feed' section with a 'Filter' button.

프로젝트 코드 원격 저장소 만들기 2

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).

Required fields are marked with an asterisk (*).

1 General

Owner *

 ChoongyeolKim

Repository name *

20260324

✔ 20260324 is available.

Great repository names are short and memorable. How about [vigilant-spoon](#)?

Description

Python Study

12 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

 Private

 Public

Anyone on the internet can see this repository. You choose who can commit.

✔  Private

You choose who can see and commit to this repository.

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

Add license

Licenses explain how others can use your code. [About licenses](#)

No license

Create repository

프로젝트 코드 원격 저장소 만들기 3

Owner *

 ChoongyeolKim

Repository name *

20260324

✓ 20260324 is available.

Great repository names are short and memorable. How about [vigilant-spoon?](#)

Description

Python Study

12 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

Choose a .gitignore template

Search templates

Packer

Perl

Phalcon

PlayFramework

Plone

Prestashop

Processing

PureScript

Python

Qooxdoo

Qt

No .gitignore

프로젝트 코드 원격 저장소 만들기 4

Create a new repository

Repositories contain a project's files and version history. Have a project elsewhere? [Import a repository](#).
Required fields are marked with an asterisk (*).

1 General

Owner *

 ChoongyeolKim ▾

Repository name *

/ 20260324

✓ 20260324 is available.

Great repository names are short and memorable. How about [vigilant-spoon](#)?

Description

Python Study

12 / 350 characters

2 Configuration

Choose visibility *

Choose who can see and commit to this repository

Private ▾

Add README

READMEs can be used as longer descriptions. [About READMEs](#)

On ☒

Add .gitignore

.gitignore tells git which files not to track. [About ignoring files](#)

Python ▾

Add license

Licenses explain how others can use your code. [About licenses](#)

No license ▾

Create repository

프로젝트 코드 원격 저장소 만들기 5

ChoongyeolKim / 20260324 

 Issues  Pull requests  Actions  Projects  Security  Insights  Settings

 20260324 Private

 Watch 0

 main  1 Branch  0 Tags



 Add file

 Code

 ChoongyeolKim Initial commit 9e9a3e8 · now  1 Commit

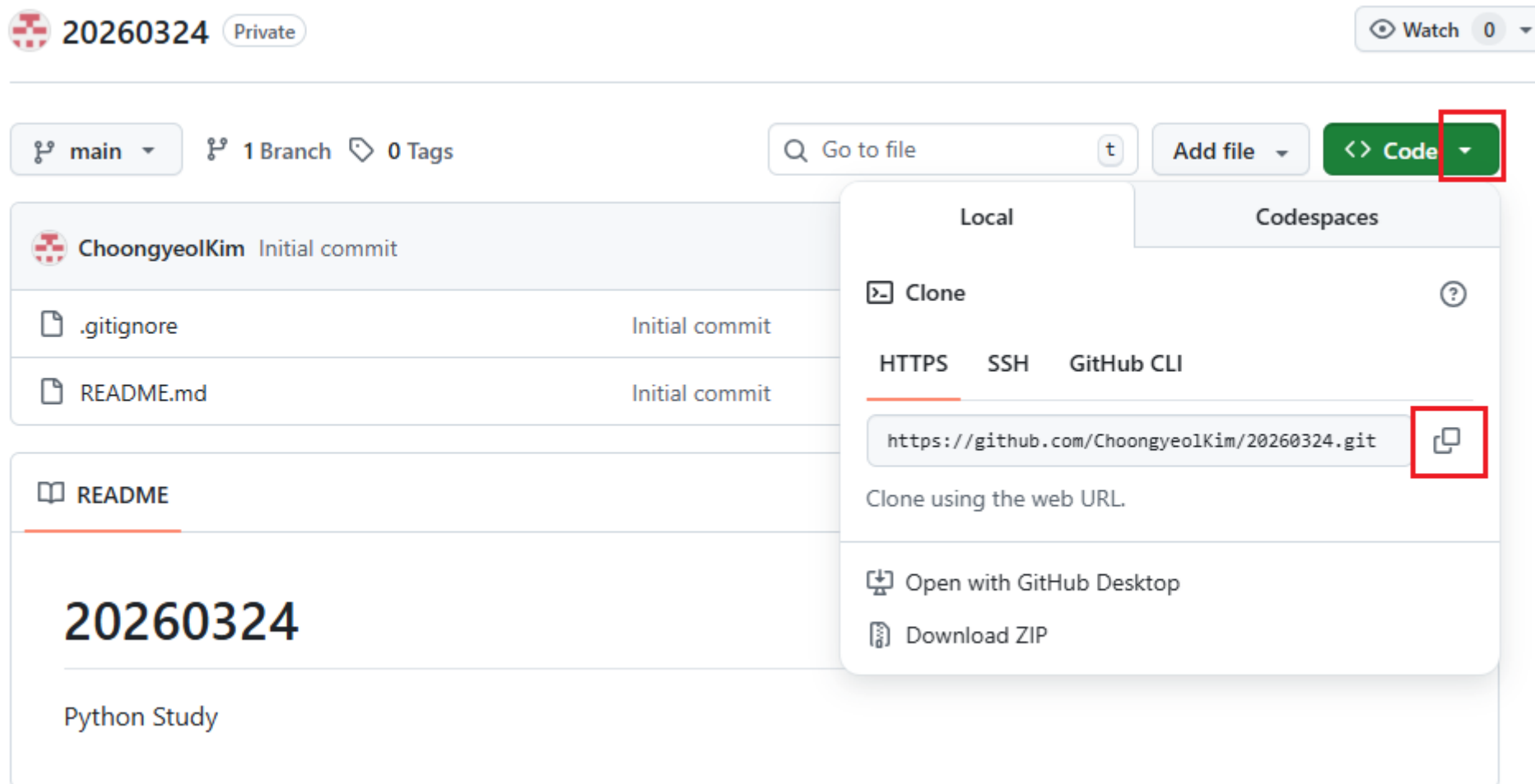
 .gitignore	Initial commit	now
 README.md	Initial commit	now

 README 

20260324

Python Study

프로젝트 코드 로컬 저장소 만들기 1



20260324 Private Watch 0

main 1 Branch 0 Tags

Go to file t Add file <> Code

ChoongyeolKim Initial commit

.gitignore Initial commit

README.md Initial commit

README

20260324

Python Study

Local Codespaces

Clone ?

HTTPS SSH GitHub CLI

https://github.com/ChoongyeolKim/20260324.git

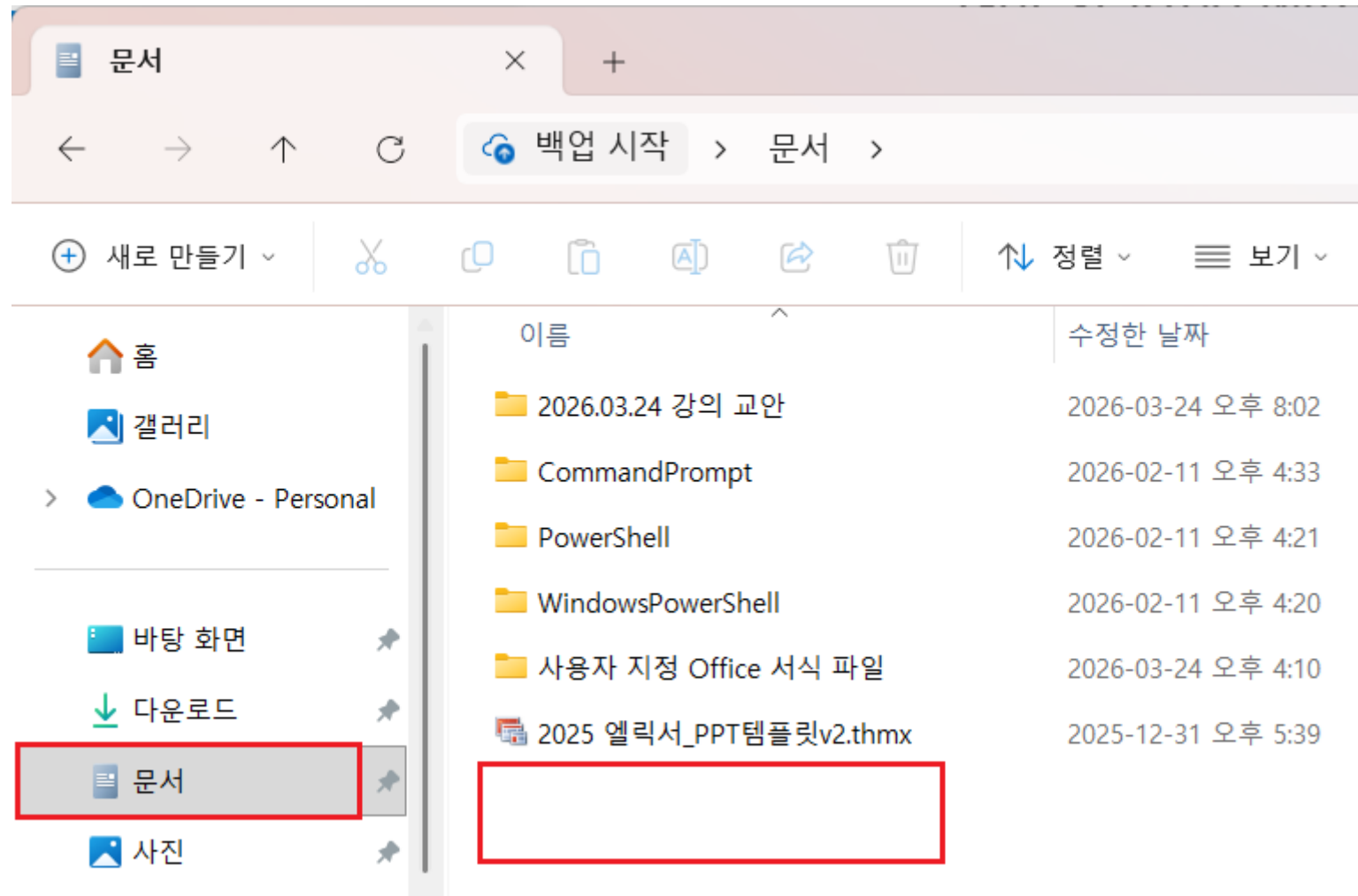
Clone using the web URL.

Open with GitHub Desktop

Download ZIP

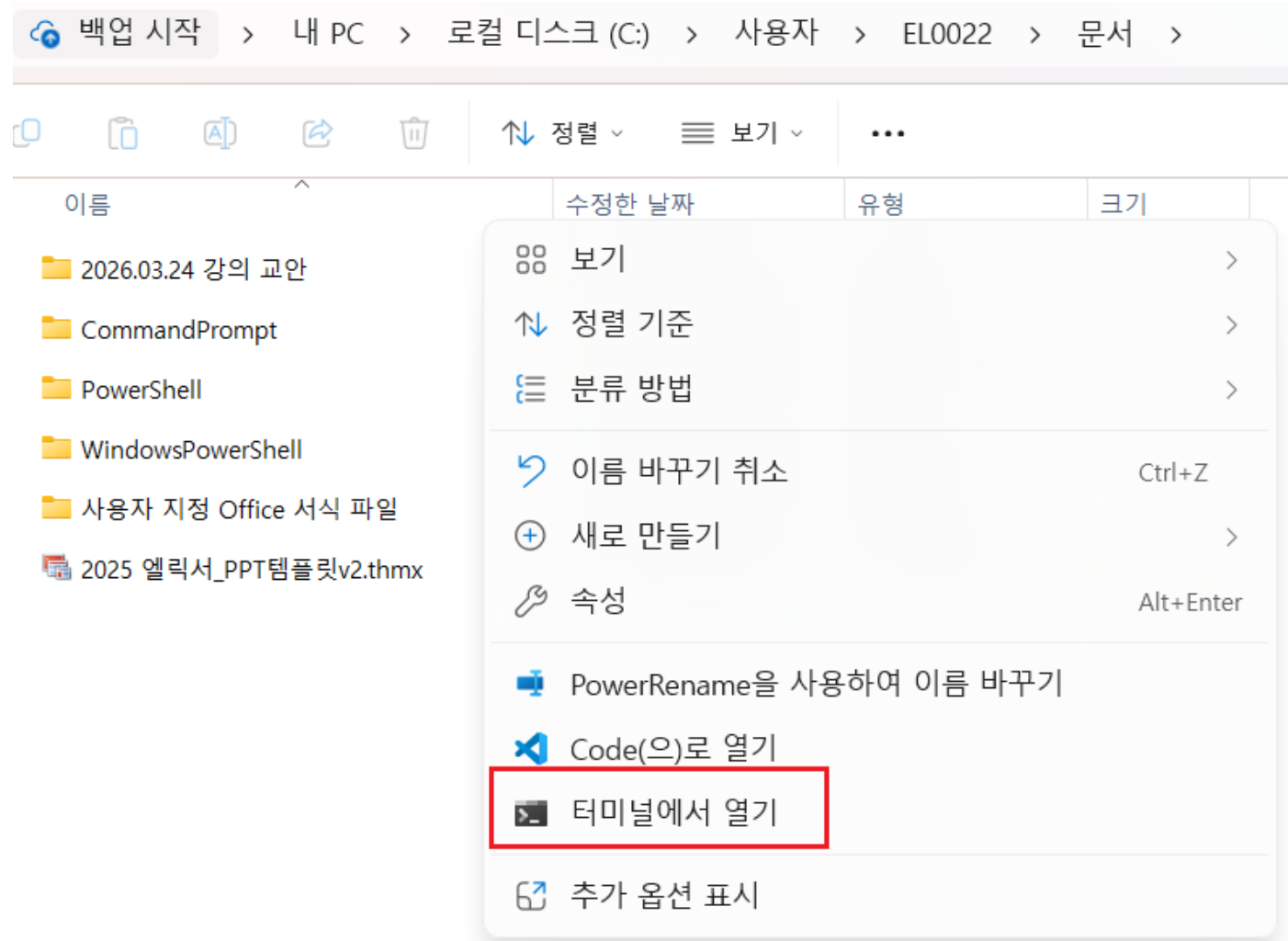
프로젝트 코드 로컬 저장소 만들기 2

C:\Users\[본인계정]\Documents



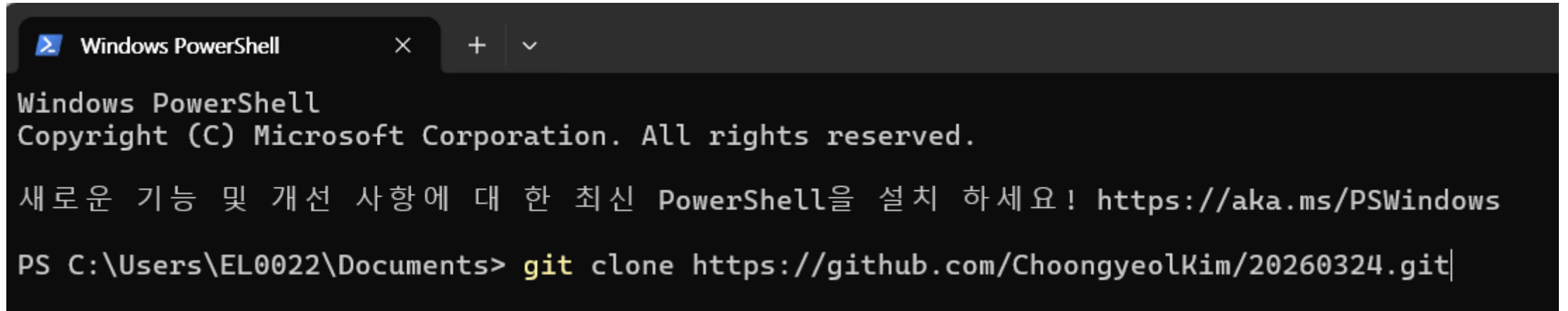
프로젝트 코드 로컬 저장소 만들기 3

마우스 우클릭 후 컨텍스트 메뉴



프로젝트 코드 로컬 저장소 만들기 4

```
git clone https://github.com/ChoongyeolKim/20260324.git
```



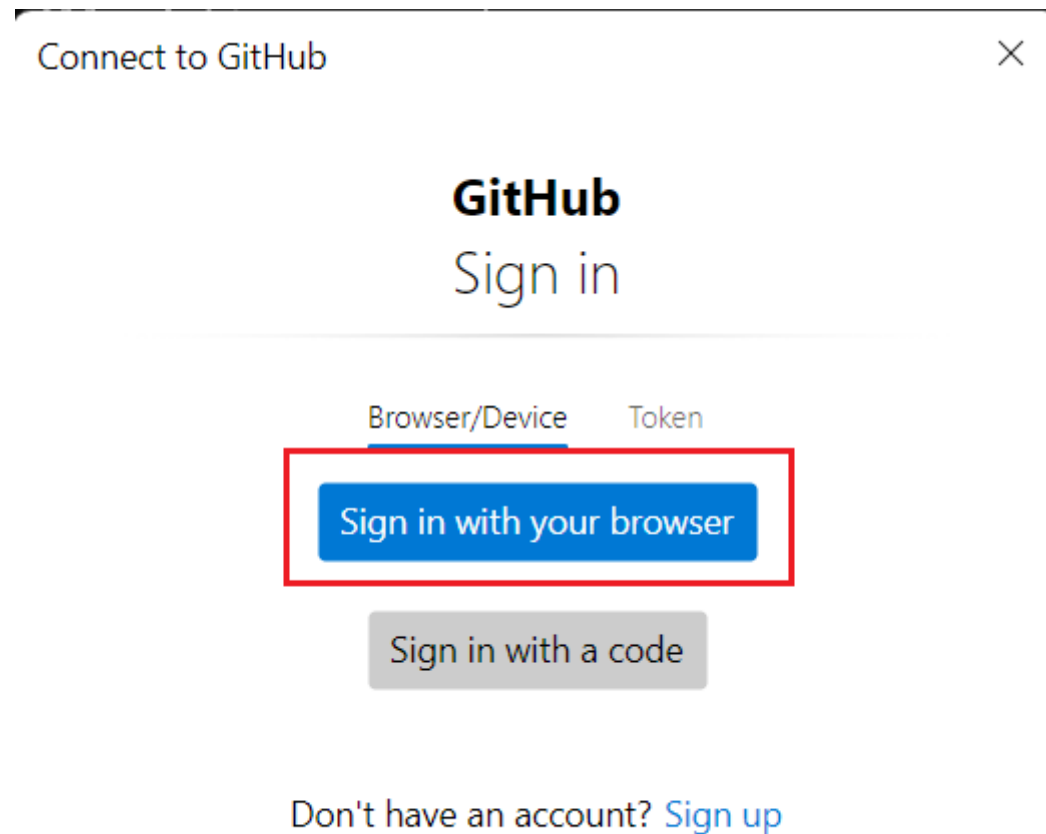
```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

새로운 기능 및 개선 사항에 대한 최신 PowerShell을 설치 하세요! https://aka.ms/PSWindows

PS C:\Users\EL0022\Documents> git clone https://github.com/ChoongyeolKim/20260324.git|
```

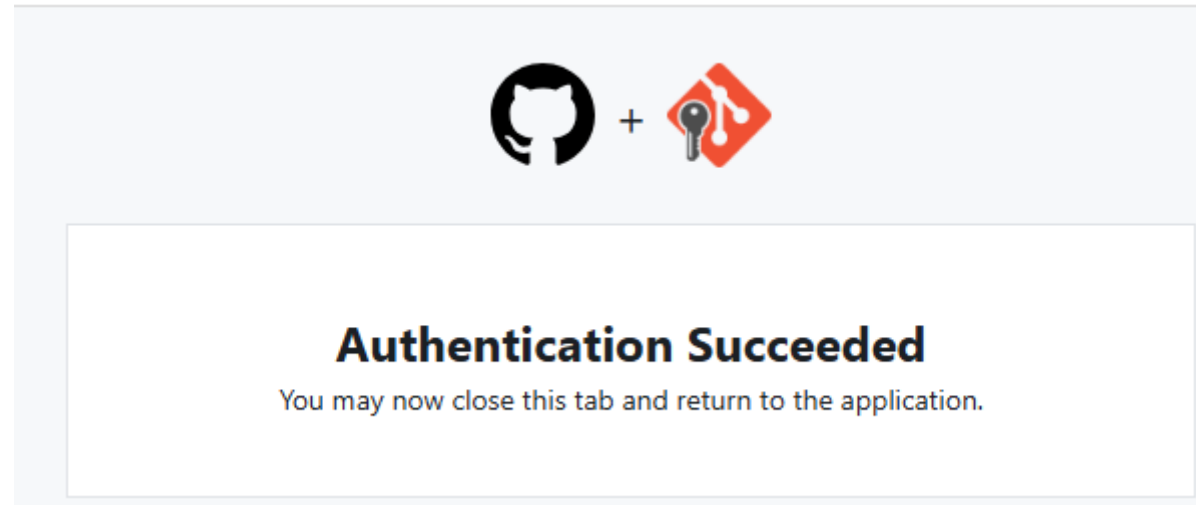
프로젝트 코드 로컬 저장소 만들기 5

원격 저장소에서 처음으로 코드를 가져오는 경우에는 다음과 같은 메뉴가 나오고
열려있는 브라우저로 로그인을 하면 완료.



프로젝트 코드 로컬 저장소 만들기 6

웹브라우저로 로그인이 성공하면 다음과 같이 메시지가 나옵니다.



프로젝트 코드 로컬 저장소 만들기 7

방금 전 설정한 github credential 은 다음과 같이 저장됩니다.

↑ [🏠](#) > [제어판](#) > [사용자 계정](#) > [자격 증명 관리자](#)

자격 증명 관리

웹 사이트, 연결된 응용 프로그램 및 네트워크에 대해 저장된 로그인 정보를 보고 삭제합니다.



웹 자격 증명



Windows 자격 증명

[자격 증명 백업\(B\)](#) [자격 증명 복원\(R\)](#)

Windows 자격 증명

[Windows 자격 증명 추가](#)

Windows 자격 증명이 없습니다.

인증서 기반 자격 증명

[인증서 기반 자격 증명 추가](#)

인증서가 없습니다.

일반 자격 증명

[일반 자격 증명 추가](#)

[git:https://github.com](https://github.com)

수정한 날짜: 오늘 ^

인터넷 또는 네트워크 주소: [git:https://github.com](https://github.com)

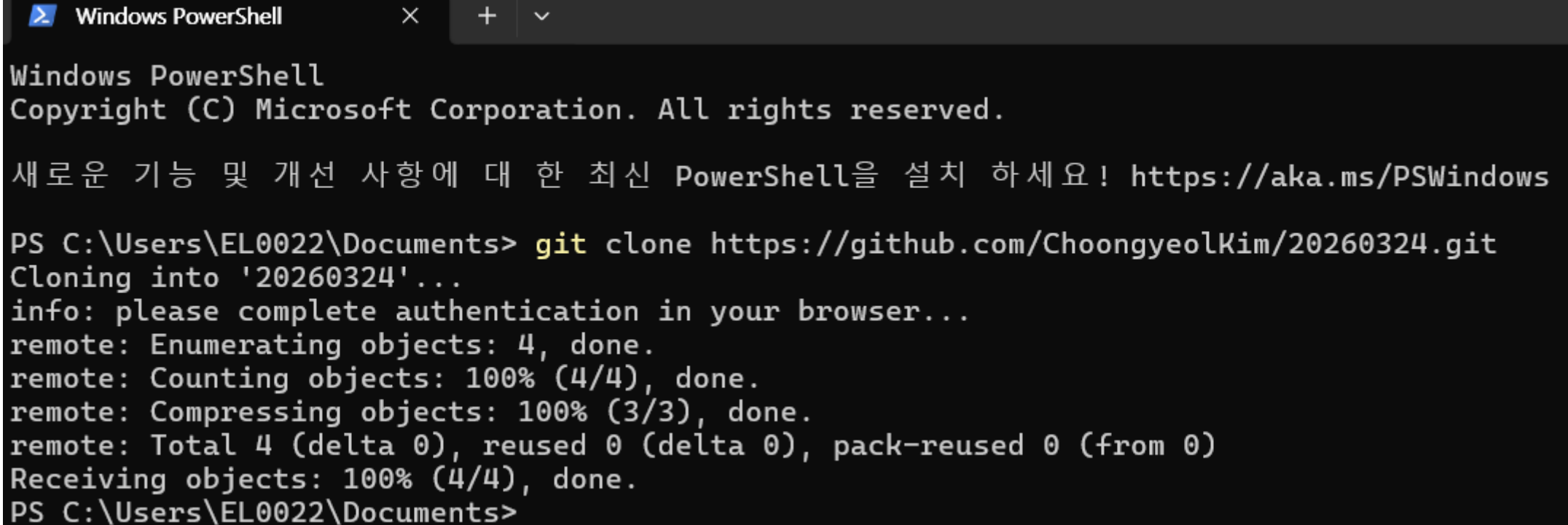
사용자 이름: ChoongyeolKim

암호:

지속성: 로컬 컴퓨터

[편집](#) [제거](#)

프로젝트 코드 로컬 저장소 만들기 8



```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

새로운 기능 및 개선 사항에 대한 최신 PowerShell을 설치 하세요! https://aka.ms/PSWindows

PS C:\Users\EL0022\Documents> git clone https://github.com/ChoongyeolKim/20260324.git
Cloning into '20260324'...
info: please complete authentication in your browser...
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Compressing objects: 100% (3/3), done.
remote: Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (4/4), done.
PS C:\Users\EL0022\Documents>
```

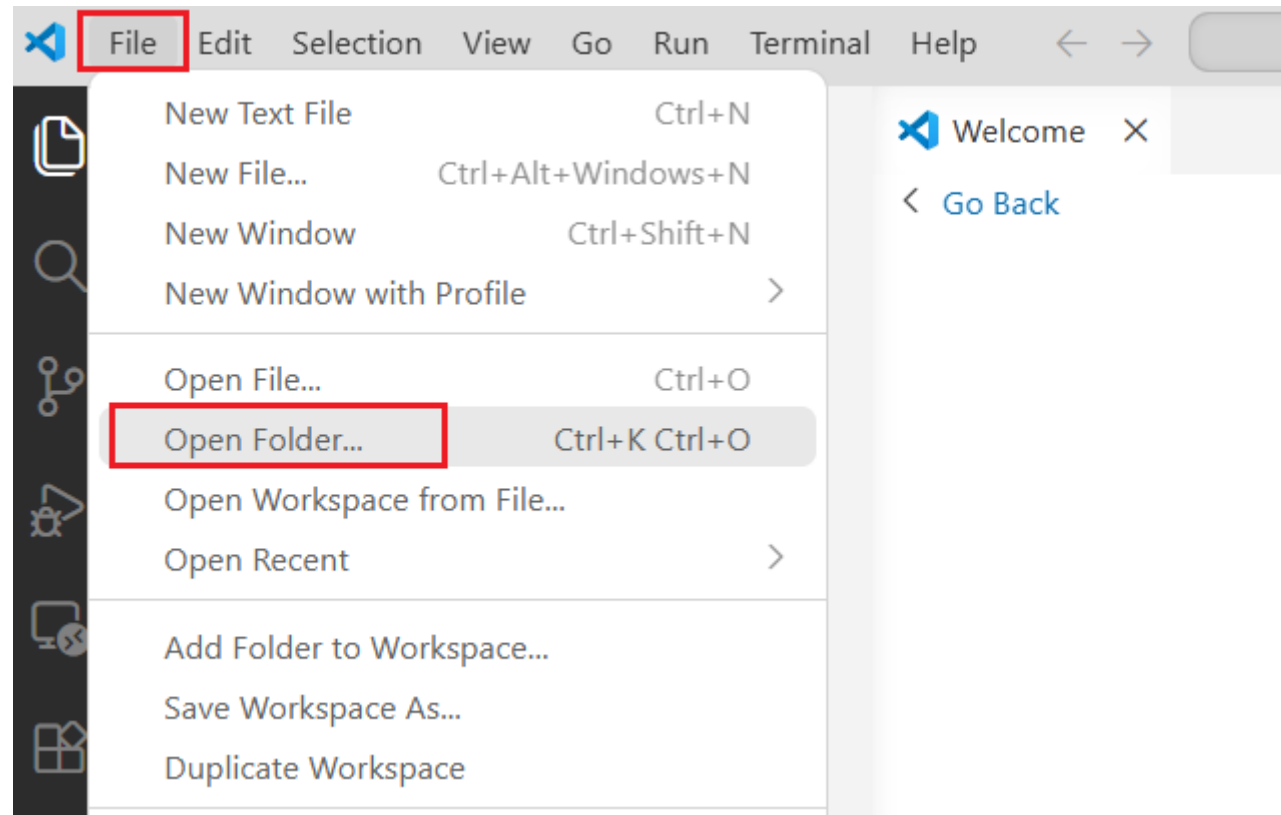
프로젝트 코드 로컬 저장소 만들기 9

이름	수정한 날짜
2026.03.24 강의 교안	2026-03-24 오후 8:02
CommandPrompt	2026-02-11 오후 4:33
PowerShell	2026-02-11 오후 4:21
WindowsPowerShell	2026-02-11 오후 4:20
사용자 지정 Office 서식 파일	2026-03-24 오후 4:10
2025 엘릭서_PPT템플릿v2.thmx	2025-12-31 오후 5:39
20260324	2026-03-24 오후 9:20

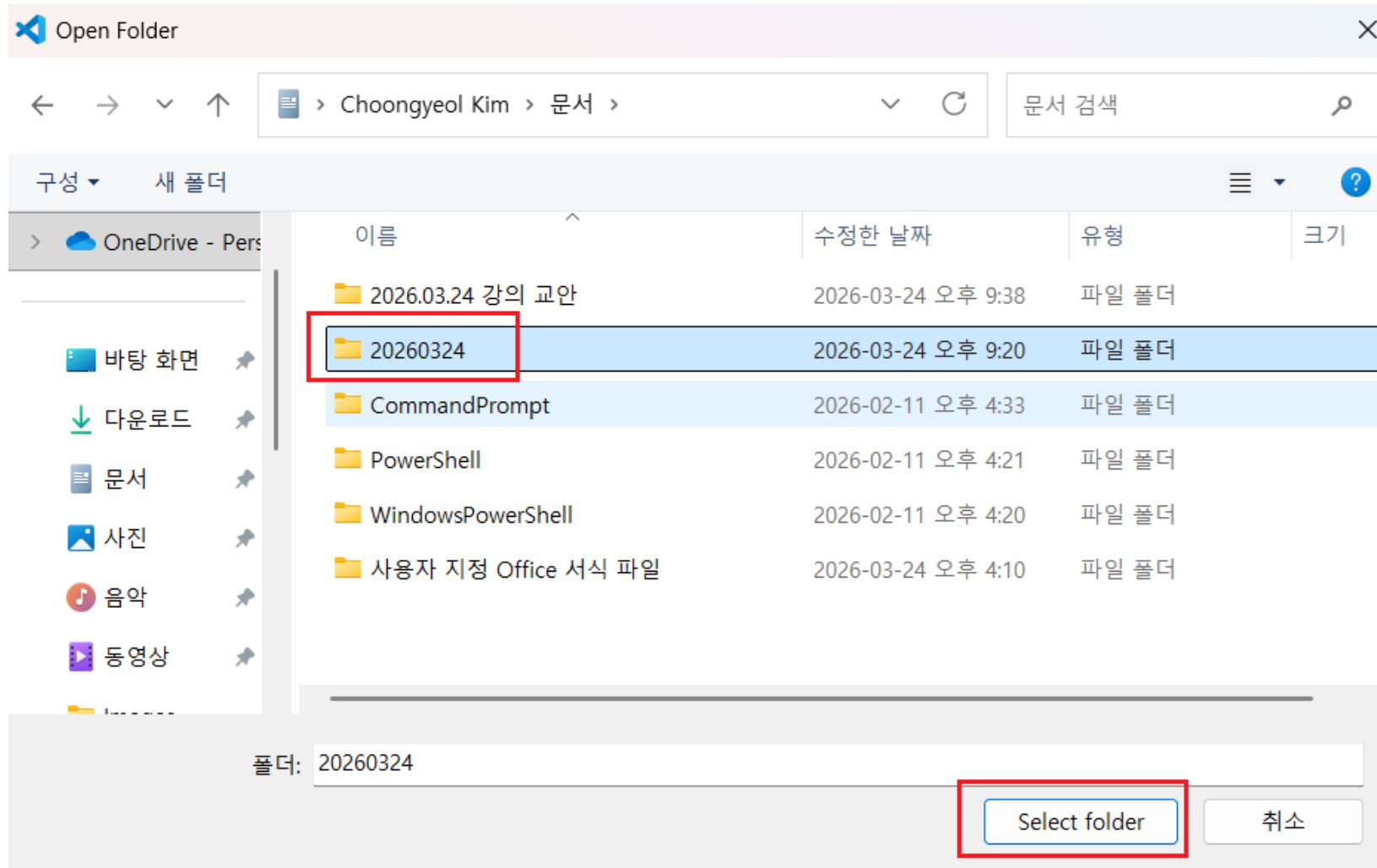
프로젝트 코드 로컬 저장소 만들기 10

```
PS C:\Users\EL0022\Documents> cd 20260324
PS C:\Users\EL0022\Documents\20260324> git config --global --list
core.editor="C:\Users\EL0022\AppData\Local\Programs\Microsoft VS Code\bin\code" --wait
PS C:\Users\EL0022\Documents\20260324> git config --local --list
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
remote.origin.url=https://github.com/ChoongyeolKim/20260324.git
remote.origin.fetch=+refs/heads/*:refs/remotes/origin/*
branch.main.remote=origin
branch.main.merge=refs/heads/main
PS C:\Users\EL0022\Documents\20260324>
```

Python 가상 환경 설정 1



Python 가상 환경 설정 2



Python 가상 환경 설정 3



Do you trust the authors of the files in this folder?

Code provides features that may automatically execute files in this folder.

If you don't trust the authors of these files, we recommend to continue in restricted mode as the files may be malicious. See [our docs](#) to learn more.

C:\Users\EL0022\Documents\20260324



Trust the authors of all files in the parent folder 'Documents'

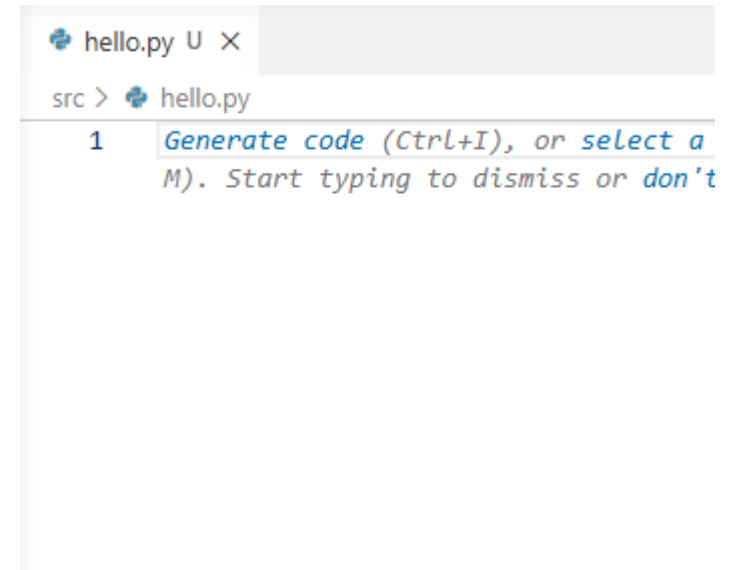
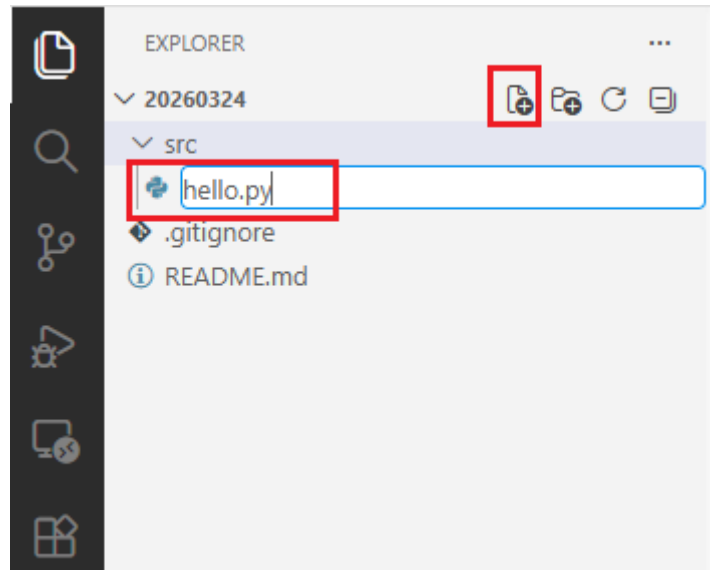
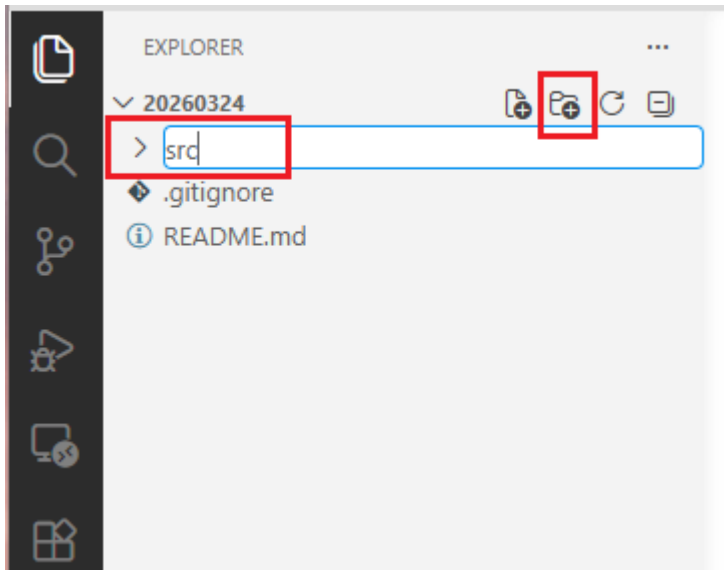
Yes, I trust the authors

Trust folder and enable all features

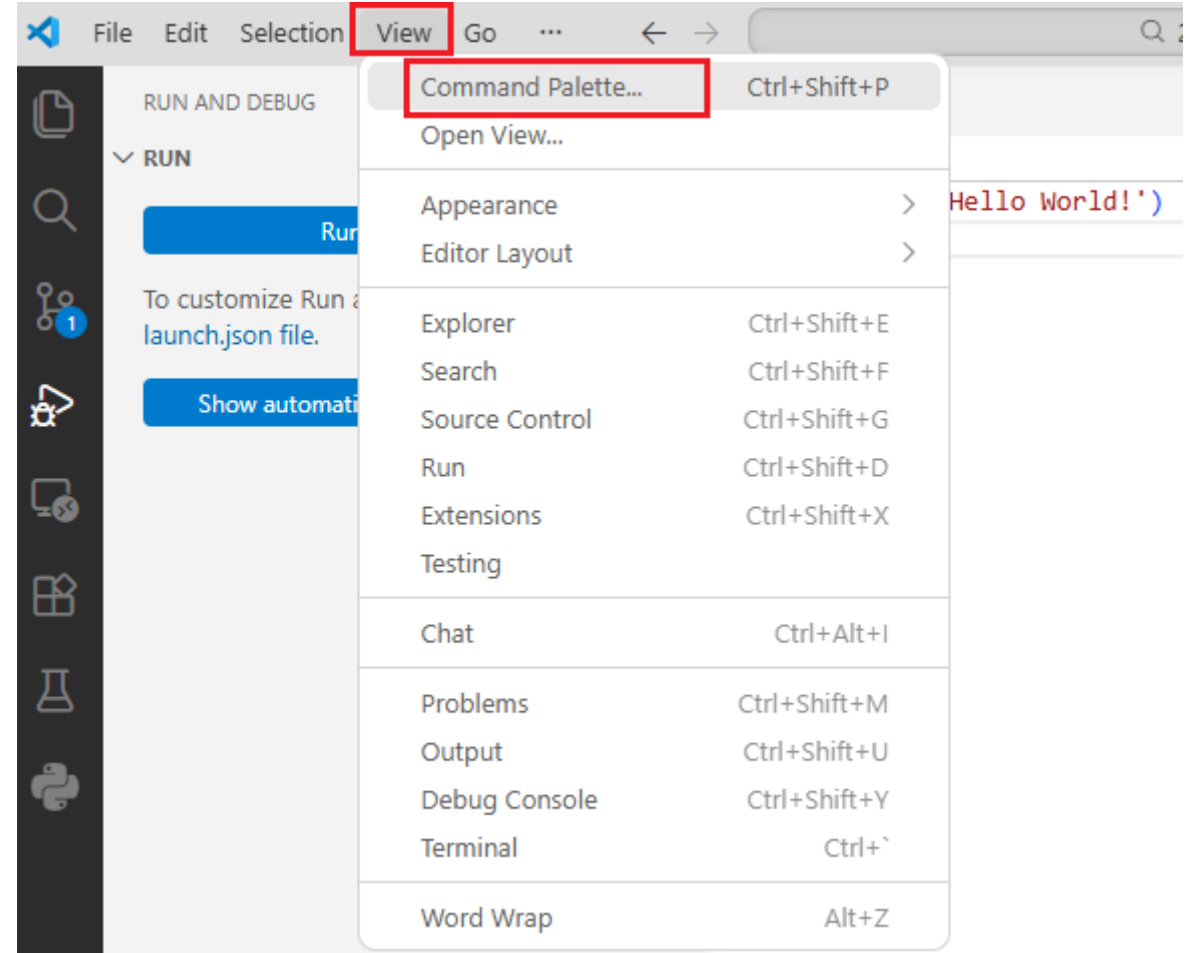
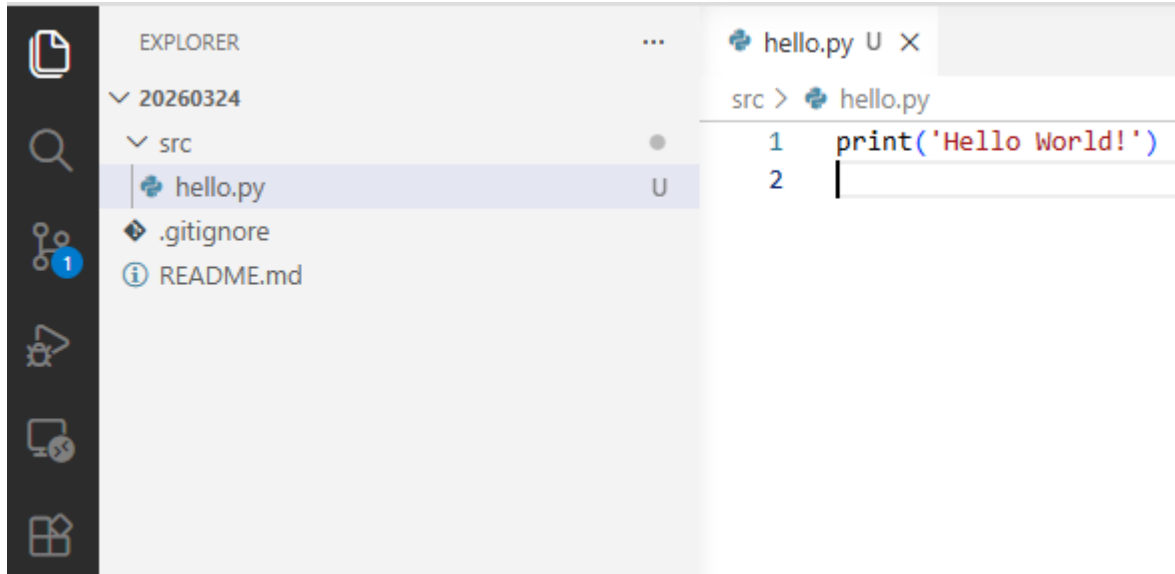
No, I don't trust the authors

Open folder in restricted mode

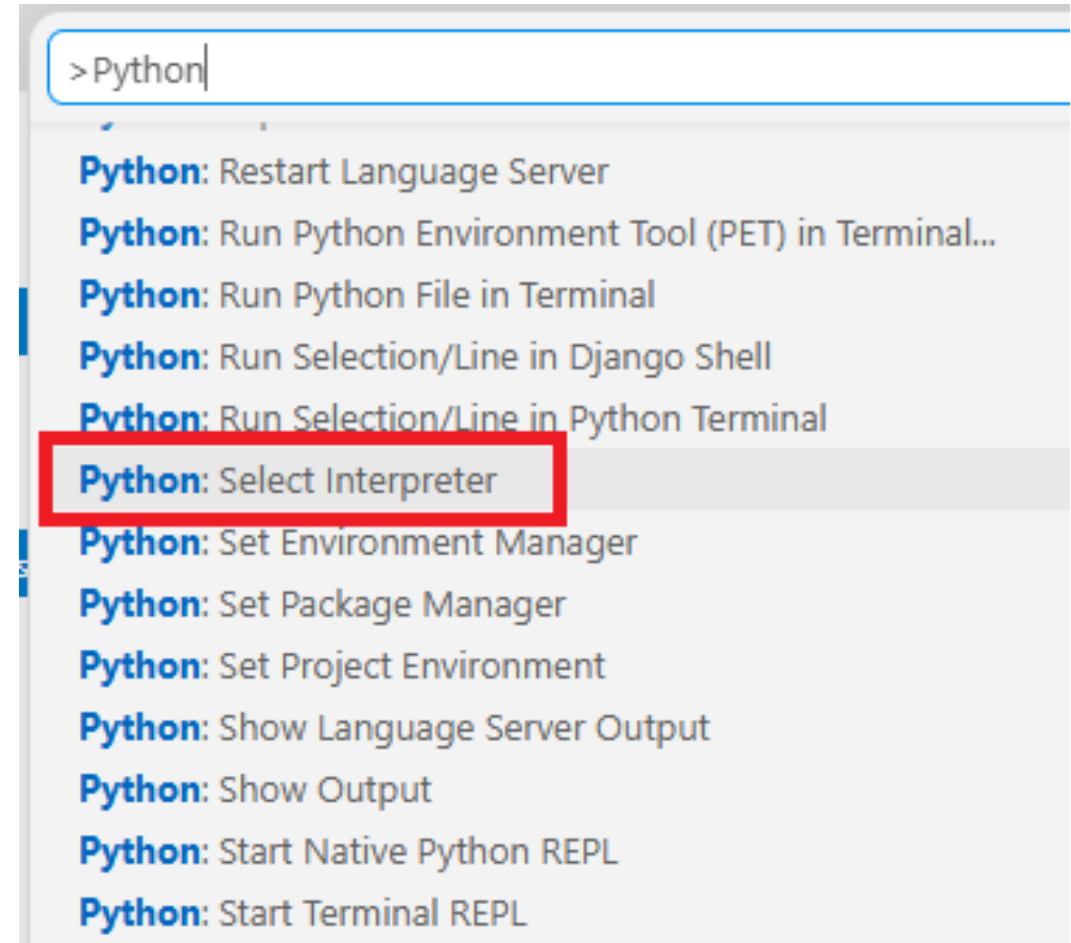
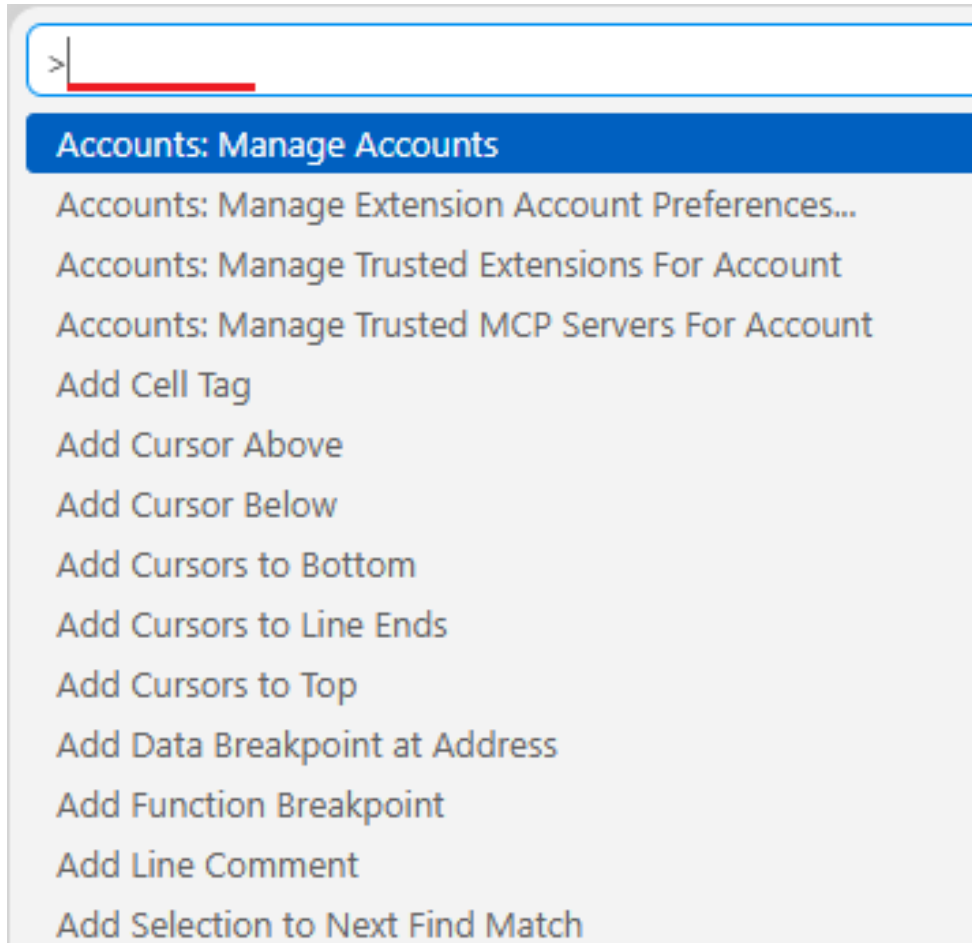
Python 가상 환경 설정 4



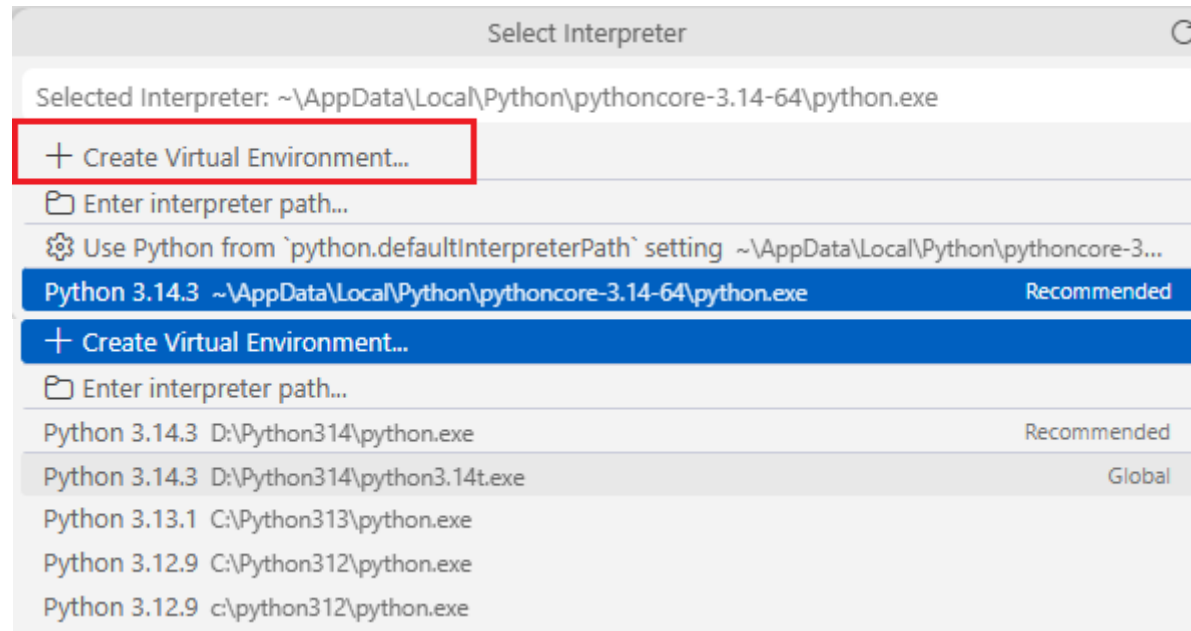
Python 가상 환경 설정 5



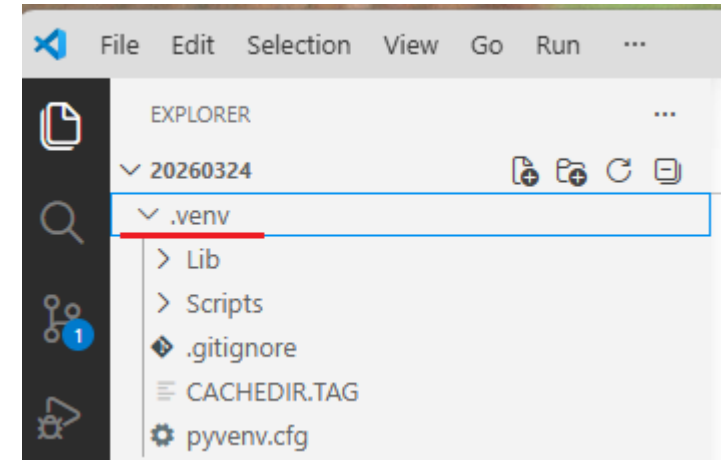
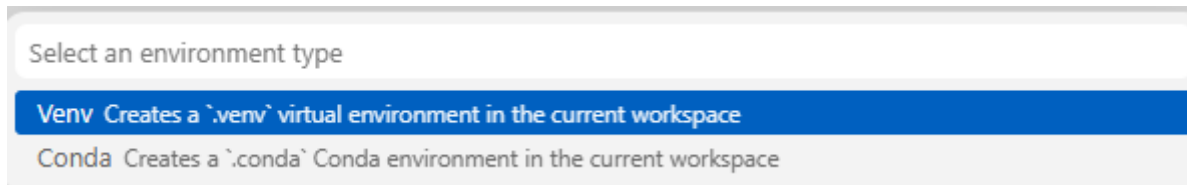
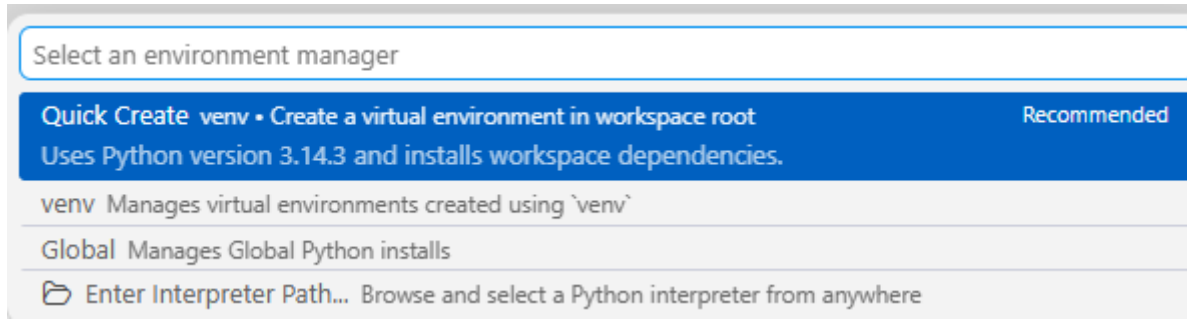
Python 가상 환경 설정 6



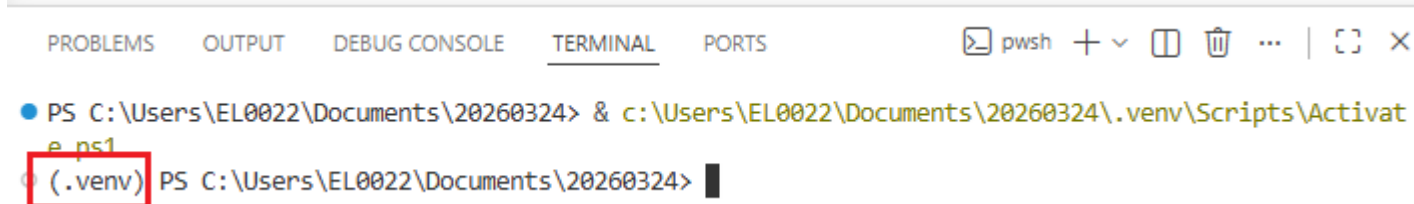
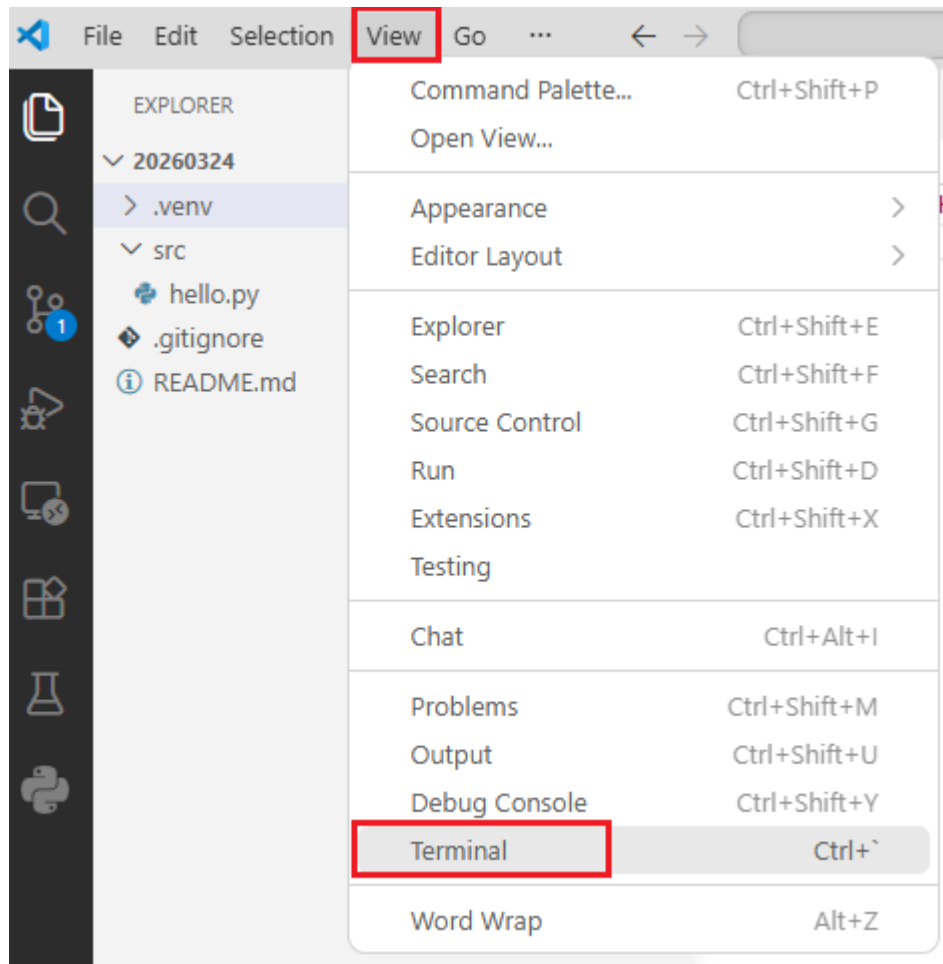
Python 가상 환경 설정 7



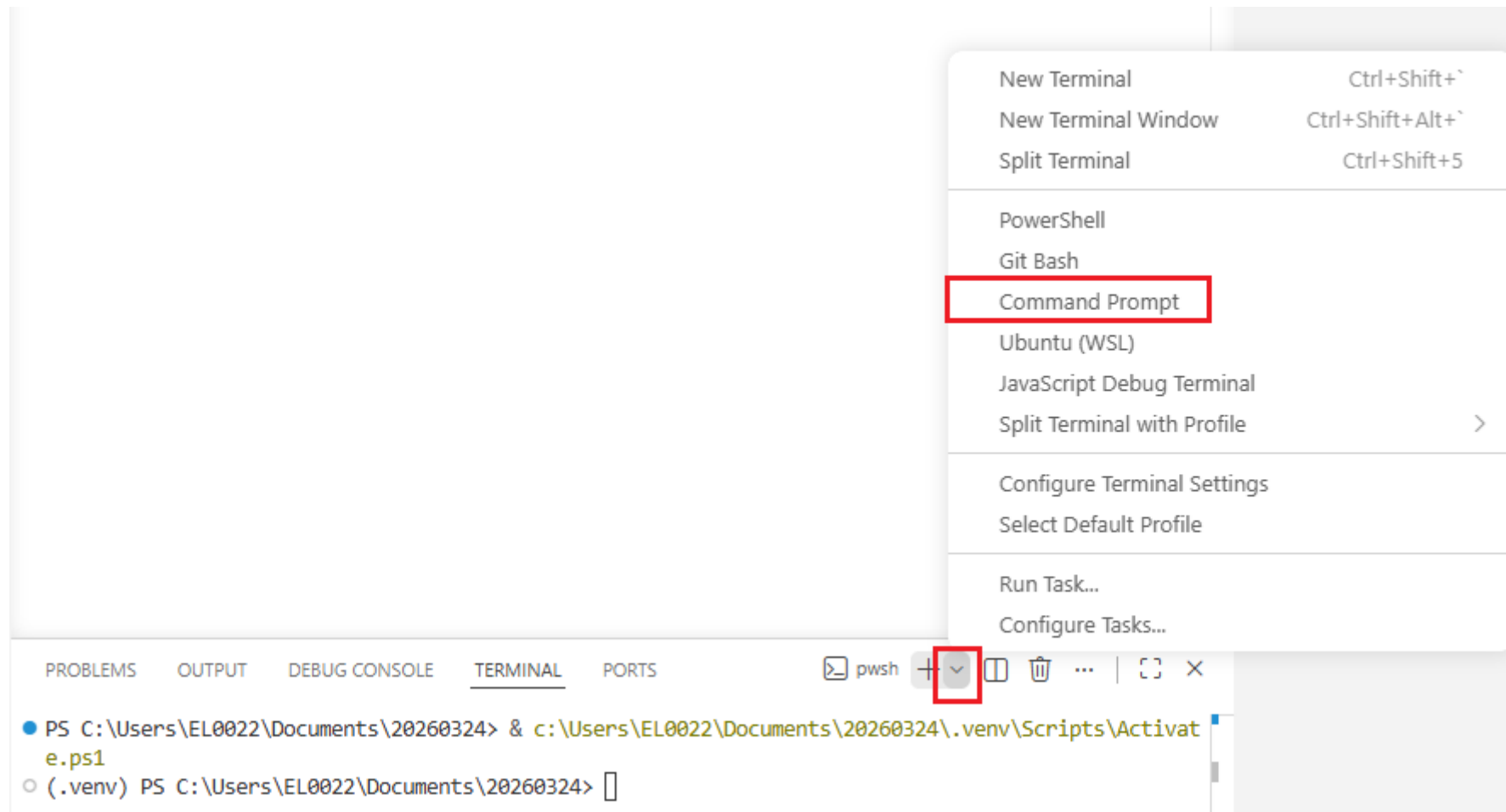
Python 가상 환경 설정 8



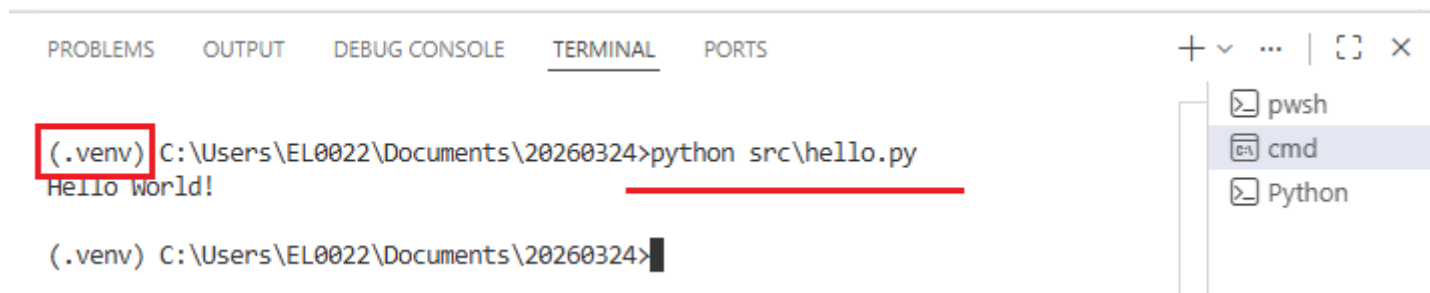
Python 파일 실행 1



Python 파일 실행 2



Python 파일 실행 3

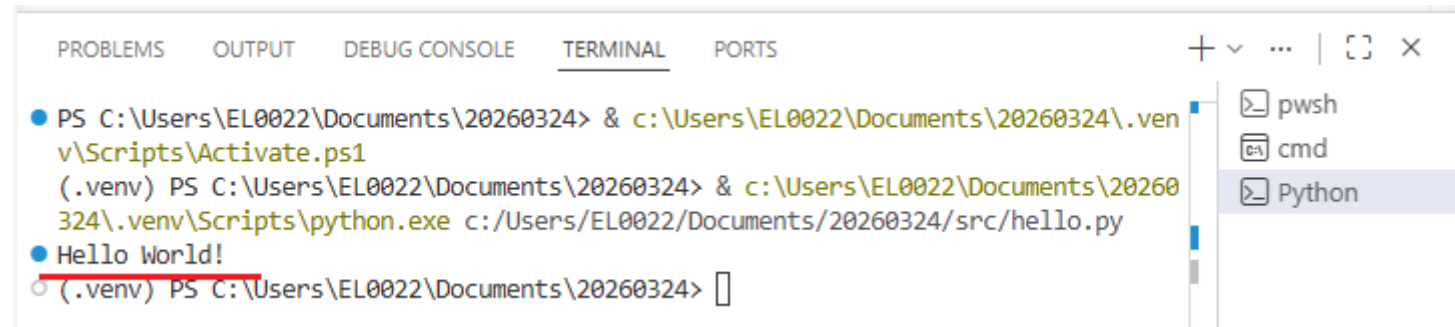
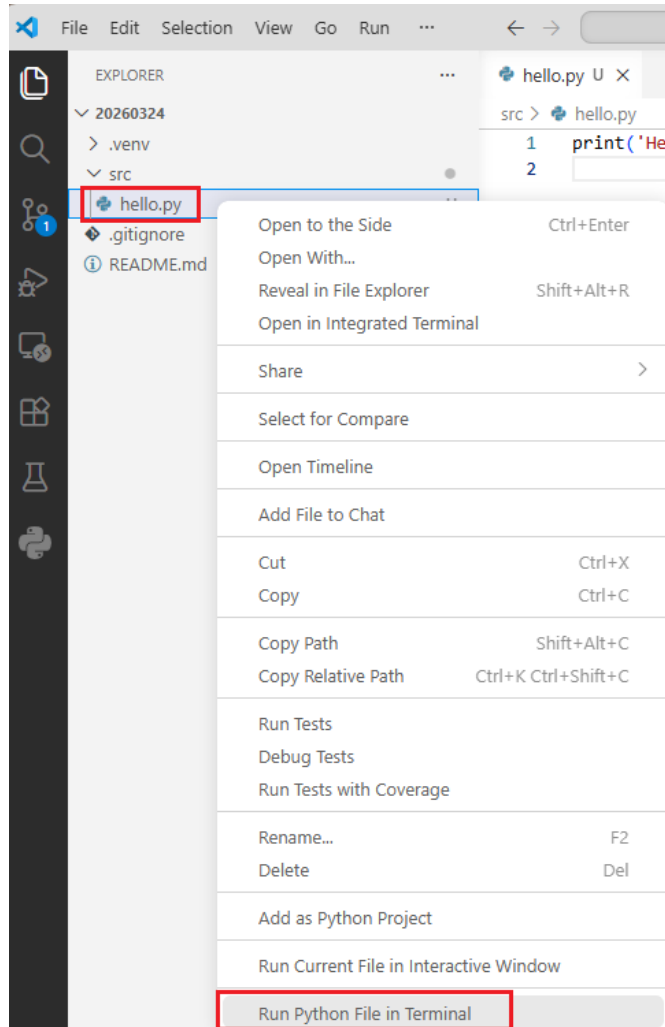


The screenshot shows a code editor's terminal window with tabs for PROBLEMS, OUTPUT, DEBUG CONSOLE, TERMINAL, and PORTS. The terminal output shows a command prompt where the user has run `python src\hello.py` in a `(.venv)` environment. The output is `Hello world!`. The `(.venv)` text is highlighted with a red box. To the right of the terminal, a dropdown menu is open, showing options: `pwsh`, `cmd` (which is selected and highlighted), and `Python`. The terminal text is as follows:

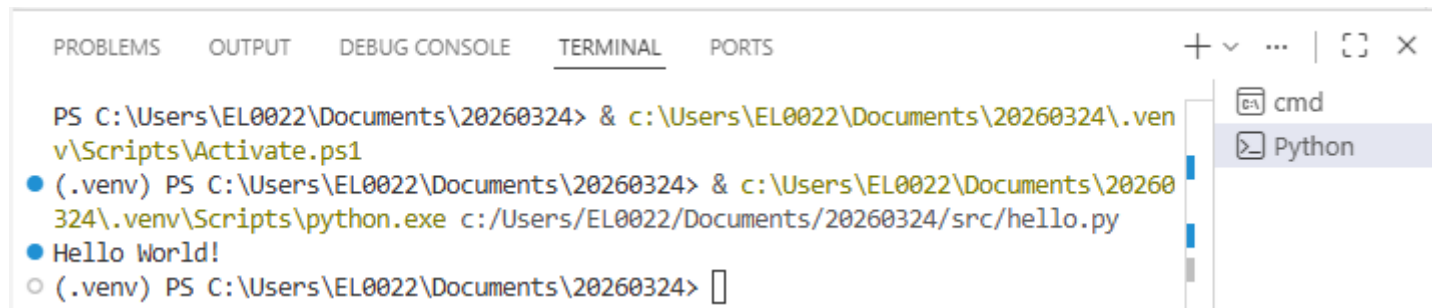
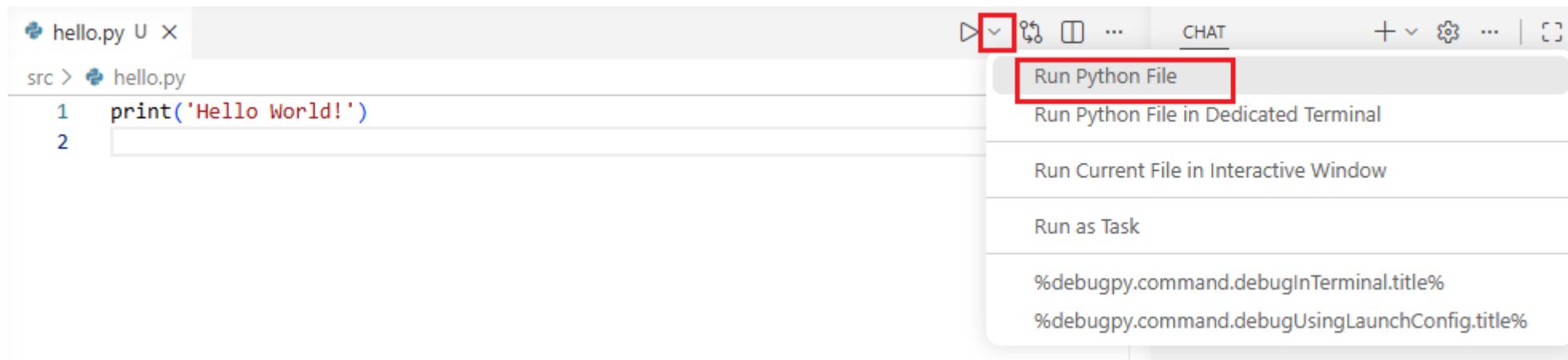
```
(.venv) C:\Users\EL0022\Documents\20260324>python src\hello.py
Hello world!

(.venv) C:\Users\EL0022\Documents\20260324>
```

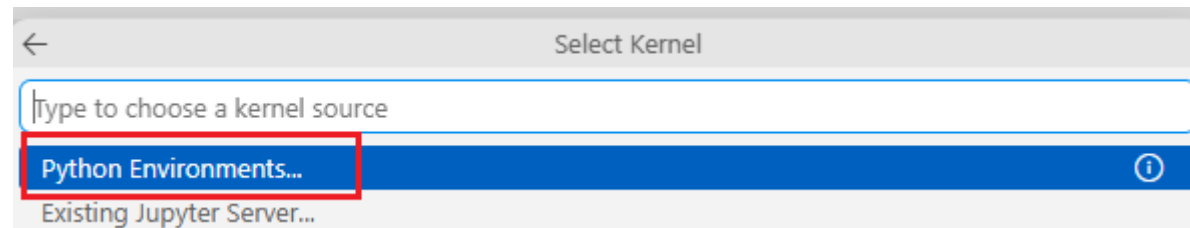
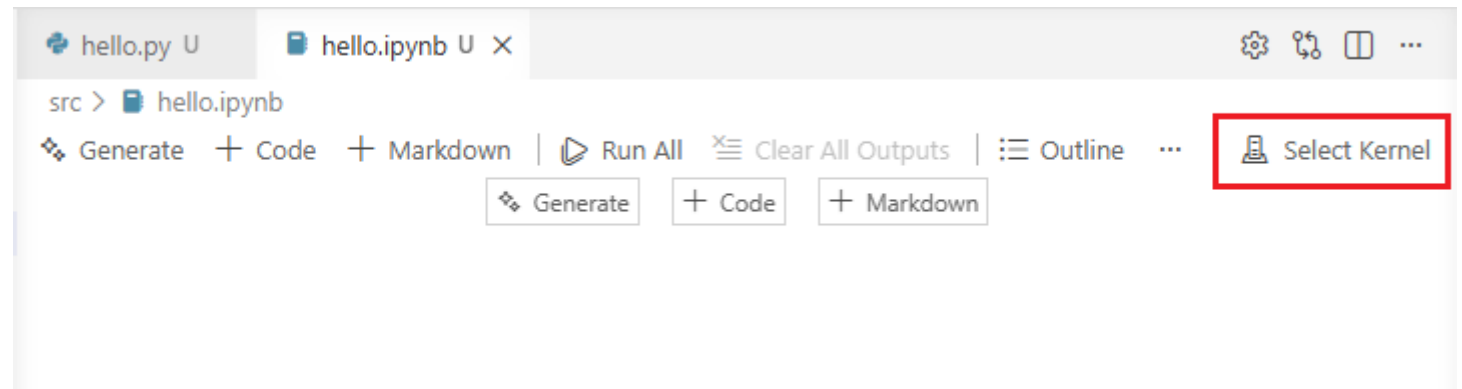
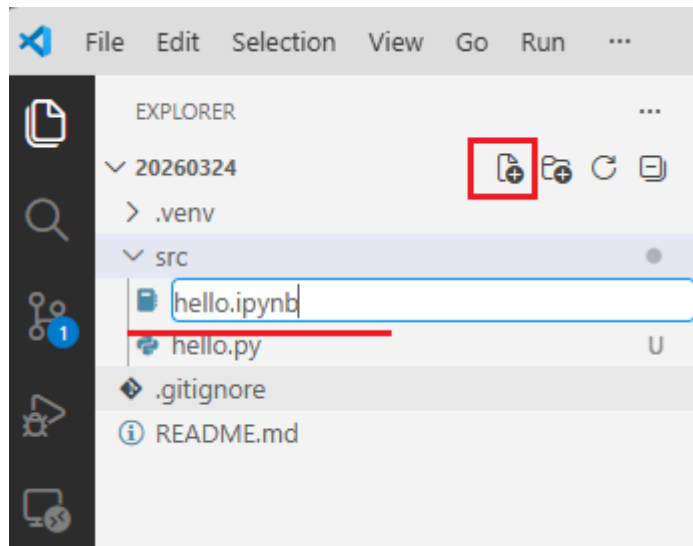

Python 파일 실행 4



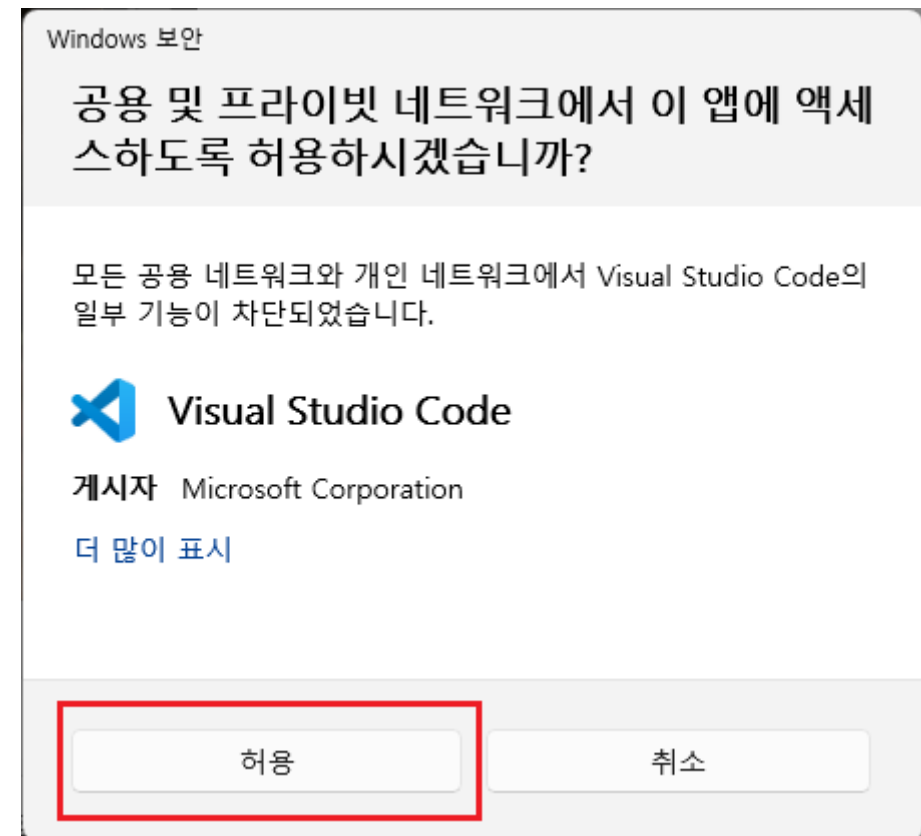
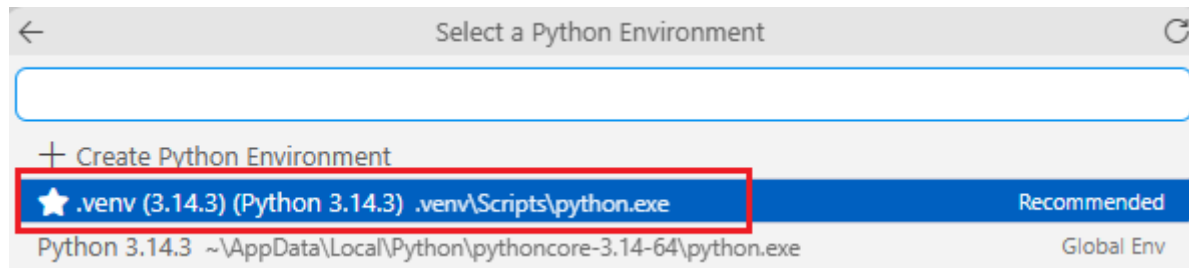
Python 파일 실행 5



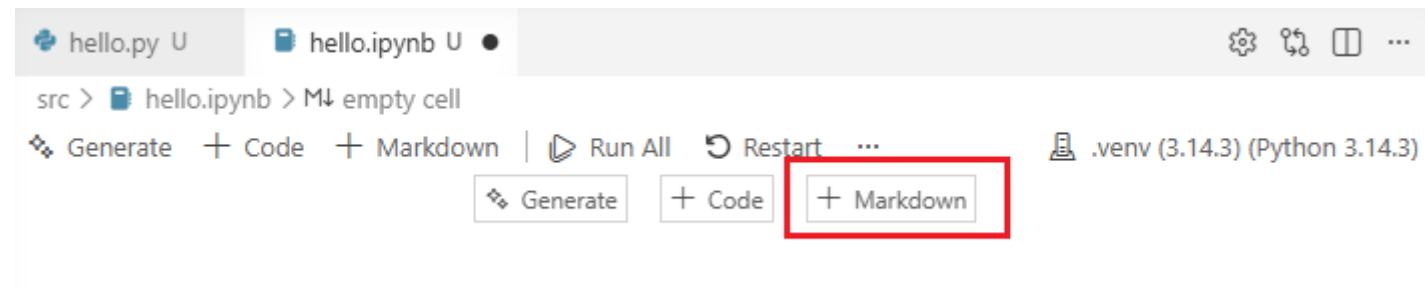
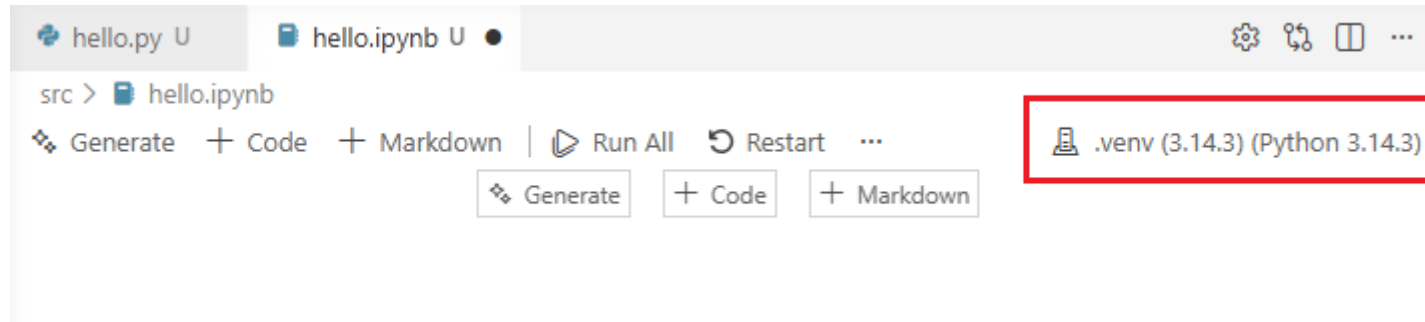
Jupyter Notebook 가상 환경 설정 1



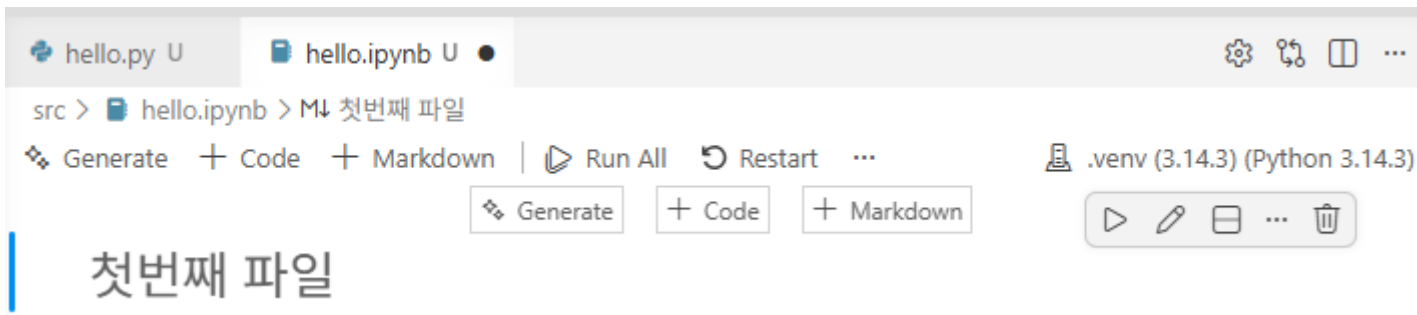
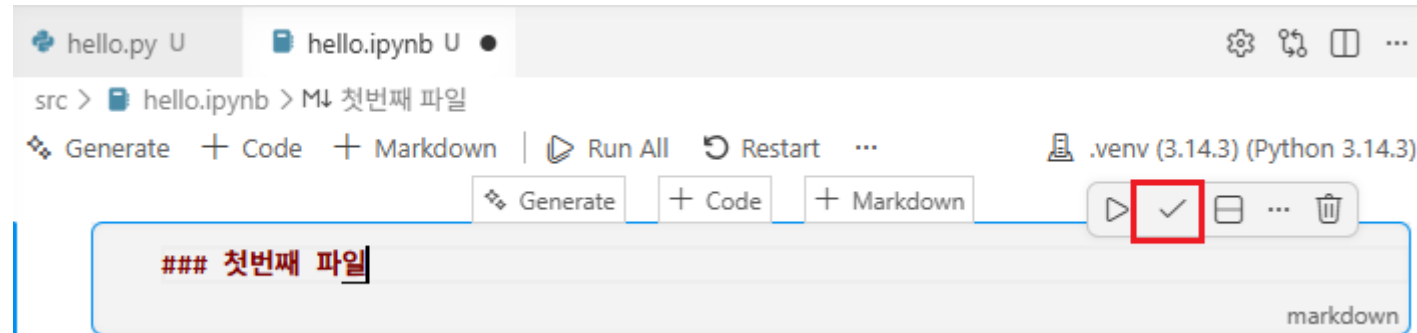
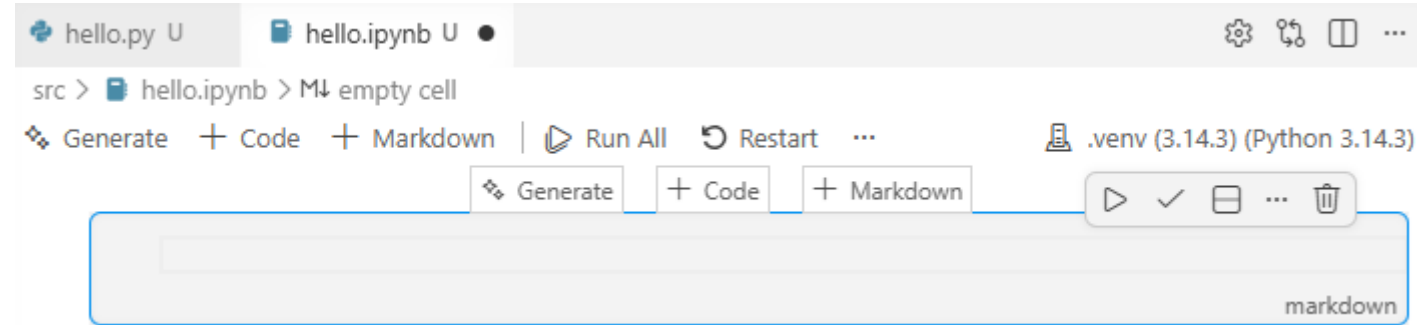
Jupyter Notebook 가상 환경 설정 2



Jupyter Notebook 파일 실행 1



Jupyter Notebook 파일 실행 2



Jupyter Notebook 파일 실행 3

다음과 같이 ipykernel 을 설치

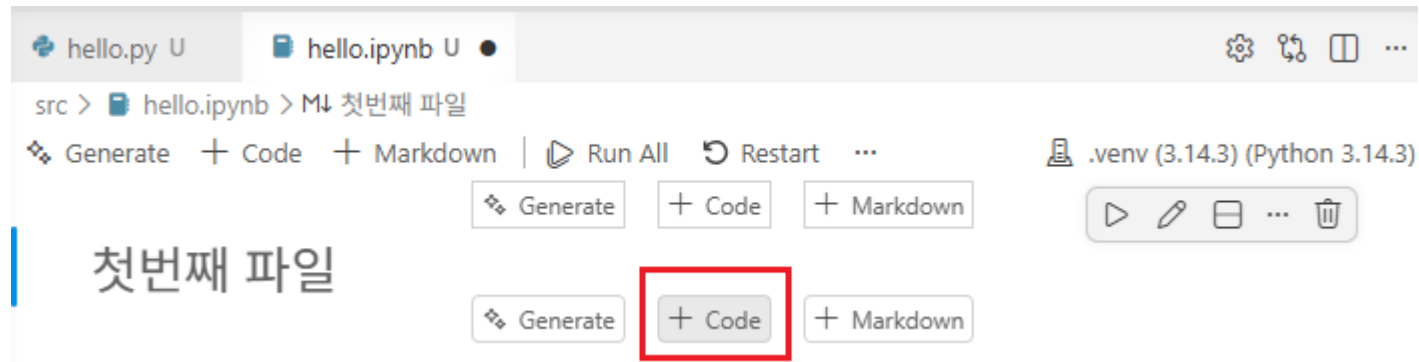
```
python -m pip install ipykernel
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS JUPYTER

```
PS C:\Users\EL0022\Documents\20260324> & c:\Users\EL0022\Documents\20260324\.venv\Scripts\Activate.ps1
(.venv) PS C:\Users\EL0022\Documents\20260324> python -m pip install --upgrade pip
Requirement already satisfied: pip in .\.venv\Lib\site-packages (26.0.1)
(.venv) PS C:\Users\EL0022\Documents\20260324> python -m pip install ipykernel
Collecting ipykernel
  Downloading ipykernel-7.2.0-py3-none-any.whl.metadata (4.5 kB)
Collecting comm<=0.1.1 (from ipykernel)
  Downloading comm-0.2.3-py3-none-any.whl.metadata (3.7 kB)
Collecting debugpy<=1.6.5 (from ipykernel)
  Downloading debugpy-1.8.20-cp314-cp314-win_amd64.whl.metadata (1.5 kB)
Collecting ipython<=7.23.1 (from ipykernel)
  Downloading ipython-9.11.0-py3-none-any.whl.metadata (4.5 kB)
```

Jupyter Notebook 파일 실행 4

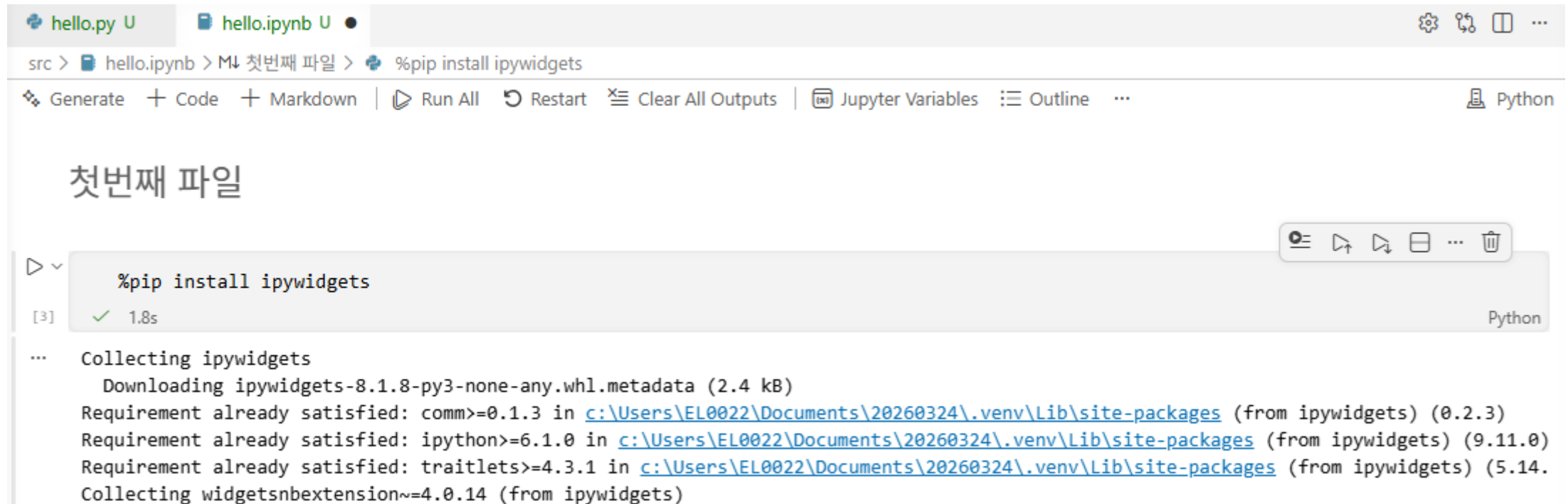
다음과 같이 코드 셀을 추가



Jupyter Notebook 파일 실행 5

다음과 같이 ipywidgets 을 설치

```
%pip install ipywidgets
```

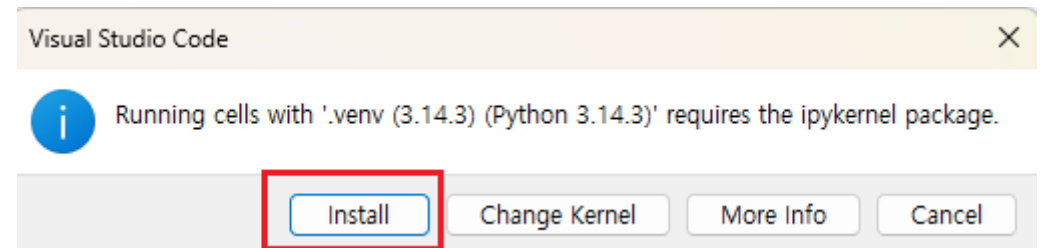
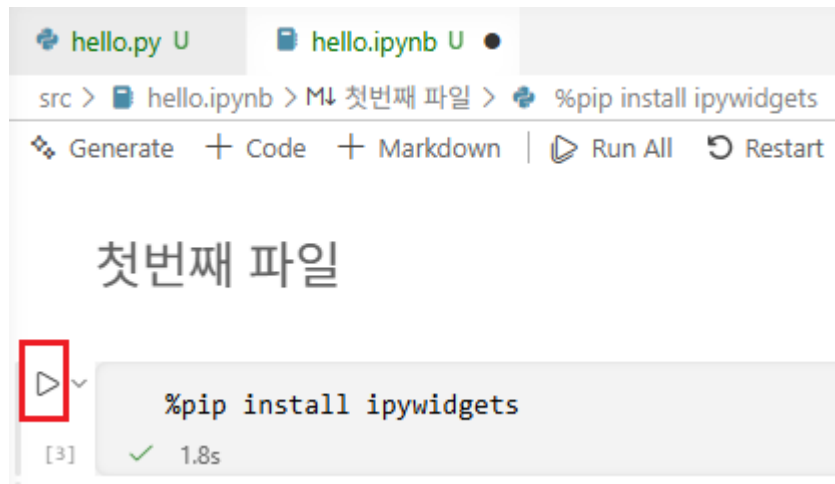


The screenshot shows a Jupyter Notebook window with two tabs: 'hello.py U' and 'hello.ipynb U'. The active tab is 'hello.ipynb U'. The notebook's file browser shows 'src > hello.ipynb > M 첫번째 파일 > %pip install ipywidgets'. The toolbar includes 'Generate', 'Code', 'Markdown', 'Run All', 'Restart', 'Clear All Outputs', 'Jupyter Variables', 'Outline', and 'Python'. The main content area is titled '첫번째 파일'. A code cell is shown with the command '%pip install ipywidgets'. The cell's status bar indicates '[3] ✓ 1.8s Python'. The output of the cell shows the following text:

```
... Collecting ipywidgets
  Downloading ipywidgets-8.1.8-py3-none-any.whl.metadata (2.4 kB)
Requirement already satisfied: comm>=0.1.3 in c:\Users\EL0022\Documents\20260324\.venv\Lib\site-packages (from ipywidgets) (0.2.3)
Requirement already satisfied: ipython>=6.1.0 in c:\Users\EL0022\Documents\20260324\.venv\Lib\site-packages (from ipywidgets) (9.11.0)
Requirement already satisfied: traitlets>=4.3.1 in c:\Users\EL0022\Documents\20260324\.venv\Lib\site-packages (from ipywidgets) (5.14.
Collecting widgetsnbextension~=4.0.14 (from ipywidgets)
```

Jupyter Notebook 파일 실행 6

다음과 같이 설치하면 자동으로 ipykernel 이 먼저 설치됩니다.



Jupyter Notebook 파일 실행 7

```
hello.py U hello.ipynb U
src > hello.ipynb > M 첫번째 파일 > %pip install ipywidgets
```

Generate + Code + Markdown | Run All Restart Clear All Outputs | Jupyter Variables Outline ...

```
Downloading jupyterlab_widgets-3.0.16-py3-none-any.whl (914 kB)
----- 0.0/914.9 kB ? eta -:-:--
...
----- 3/3 [ipywidgets]
```

Successfully installed ipywidgets-8.1.8 jupyterlab_widgets-3.0.16 widgetsnbextension-4.0.15
Note: you may need to restart the kernel to use updated packages.
Output is truncated. View as a [scrollable element](#) or open in a [text editor](#). Adjust cell output [settings](#)...

Generate + Code + Markdown

```
print('Hello World!')
```

[]

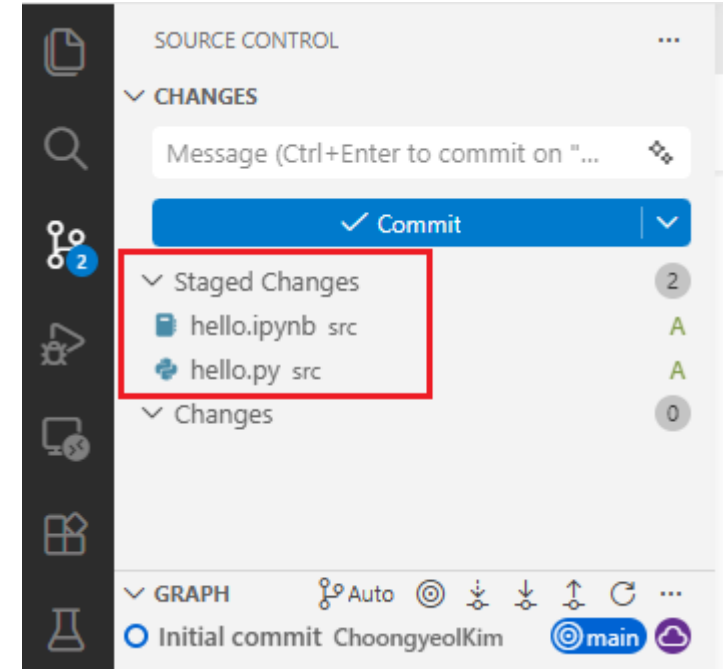
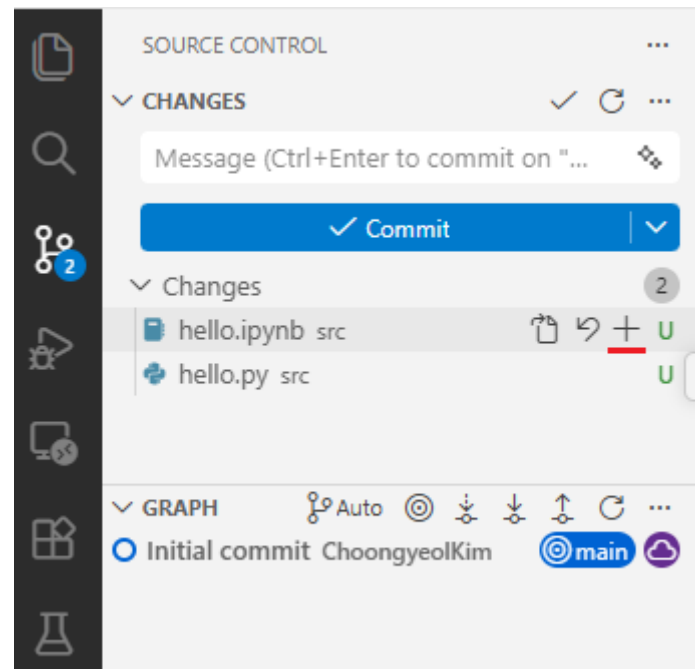
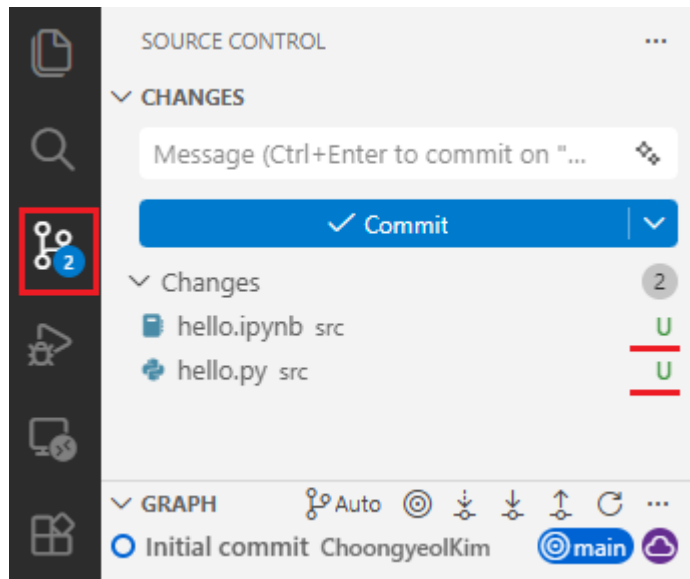
```
print('Hello World!')
```

[4] ✓ 0.0s

... Hello World!

Git 로컬 저장소에 commit 1

파일 오른쪽에 새로 추가된 파일이라는 U 표시가 있습니다.
 파일 위로 마우스를 가져가면 + 아이콘가 나타납니다. (로컬 저장소에 파일 추가 메뉴)
 지금까지 만든 2개의 파일을 모두 추가시켜 줍니다.



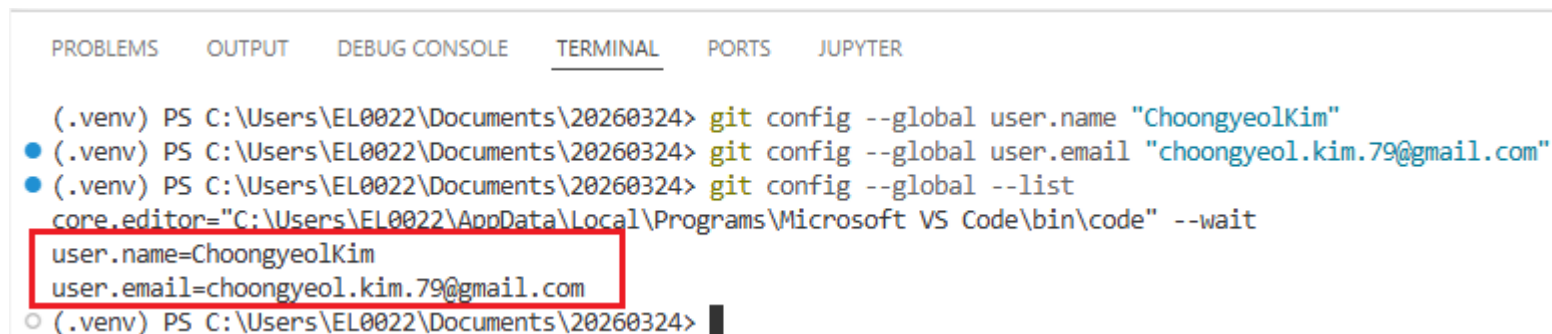
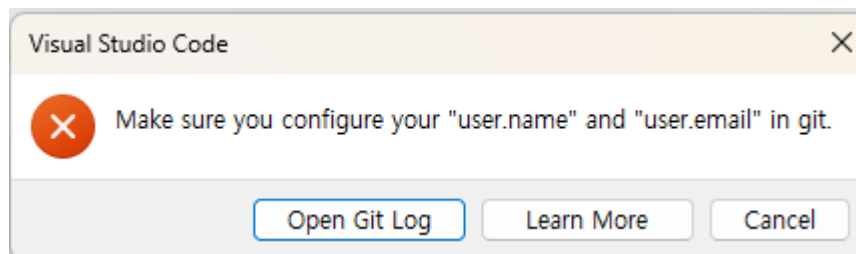
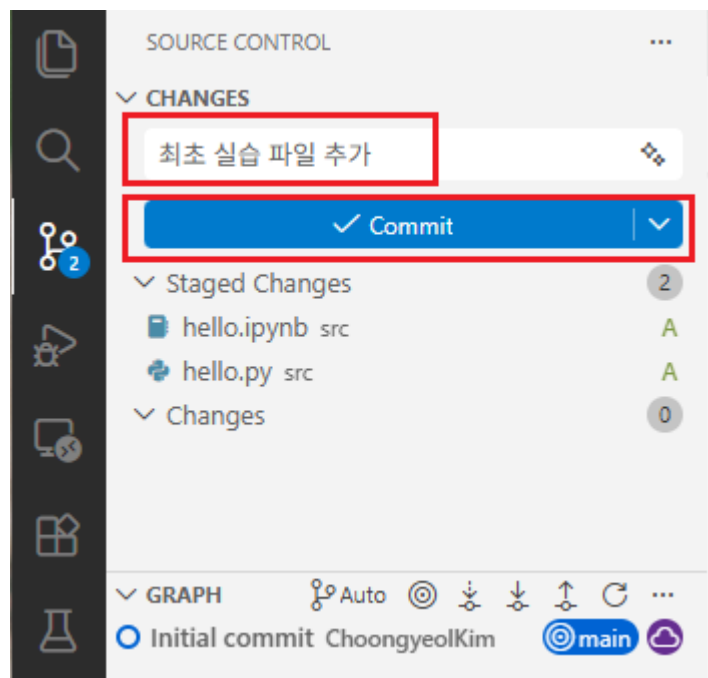
Git 로컬 저장소에 commit 2

Commit 메시지를 작성하고 commit 을 해봅니다.

다음과 같이 에러가 나면 user.name 과 user.email 을 설정합니다.

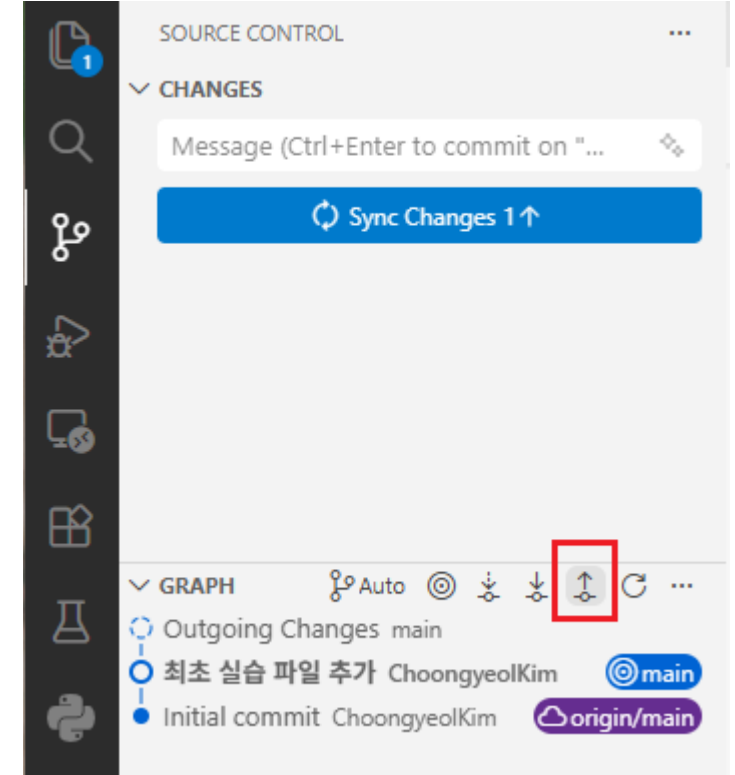
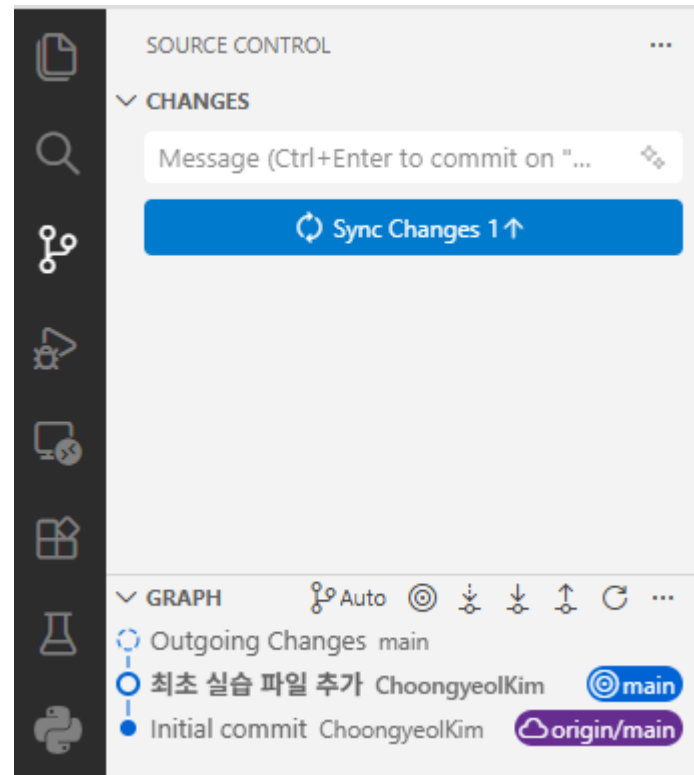
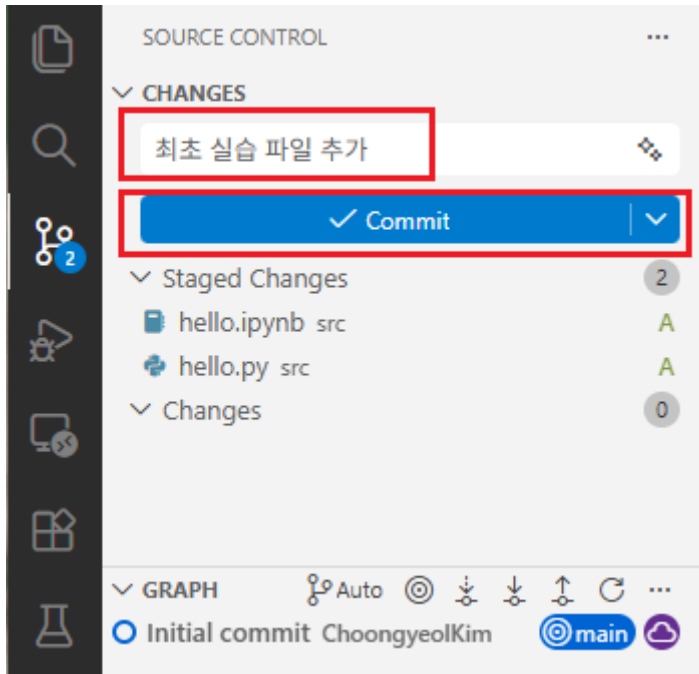
git config --global user.name "[본인 깃허브 아이디]"

git config --global user.email "[본인 깃허브 이메일]"



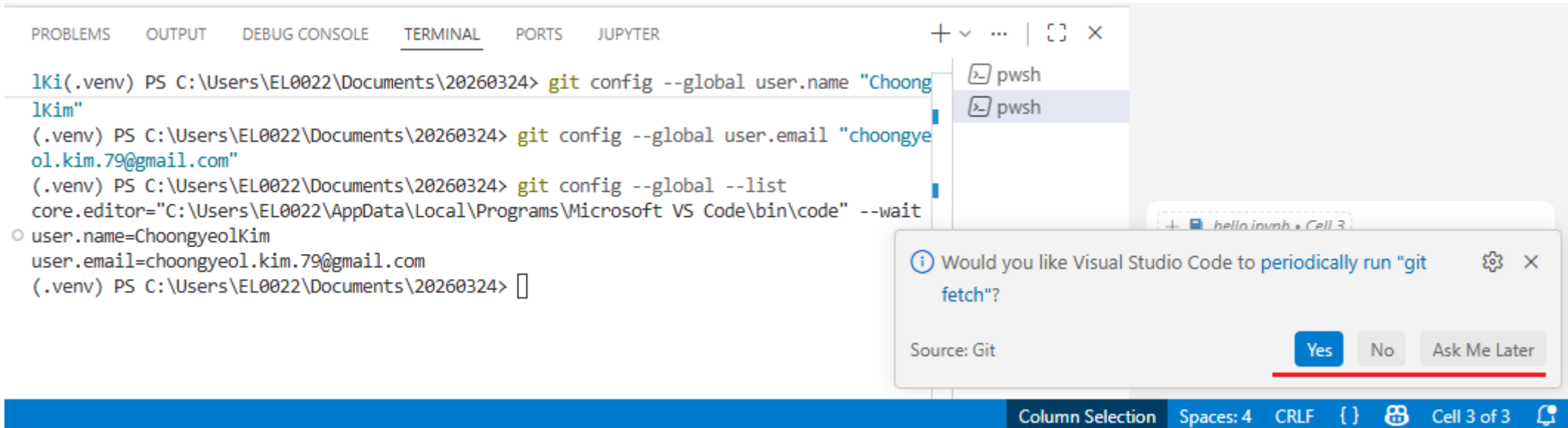
Git 원격 저장소에 push 1

다시 commit 을 시도해 보고 성공하면 원격 저장소로 push 를 해봅니다.



Git 원격 저장소에 push 2

선호하는 취향대로 Yes 혹은 No 를 선택합니다.



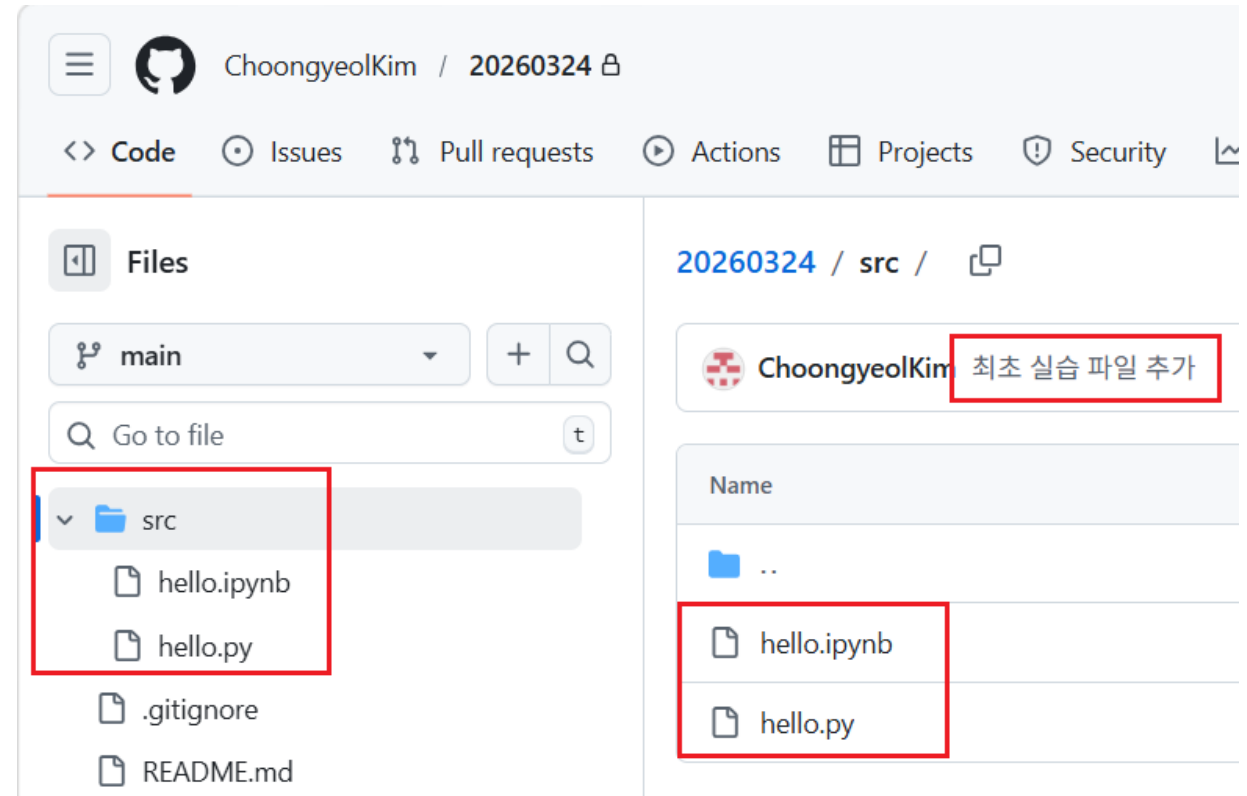
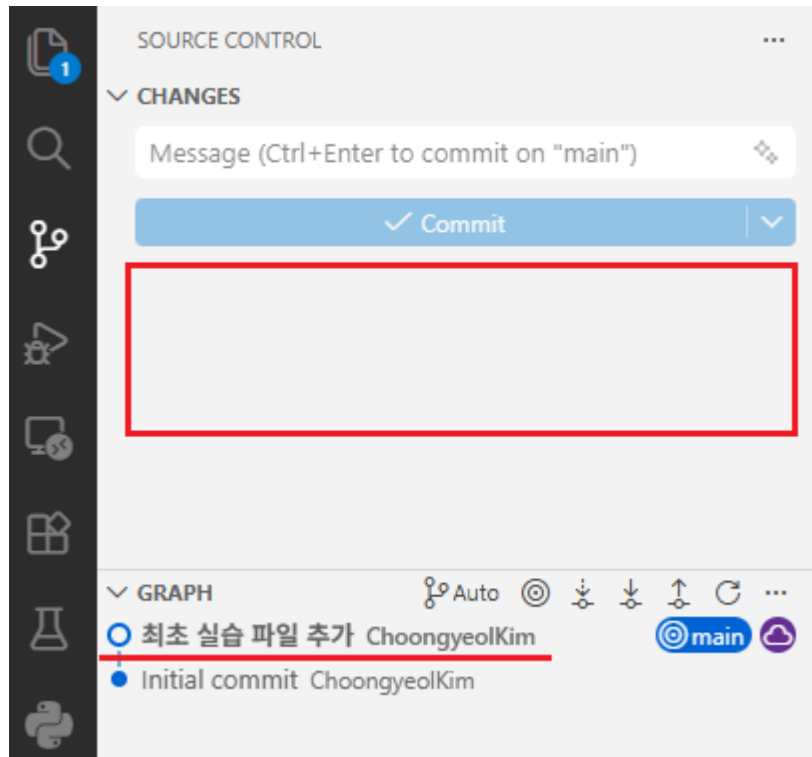
The screenshot shows the Visual Studio Code interface. The terminal window displays the following commands and output:

```
lKi(.venv) PS C:\Users\EL0022\Documents\20260324> git config --global user.name "ChoongyeolKim"
(.venv) PS C:\Users\EL0022\Documents\20260324> git config --global user.email "choongyeol.kim.79@gmail.com"
(.venv) PS C:\Users\EL0022\Documents\20260324> git config --global --list
core.editor="C:\Users\EL0022\AppData\Local\Programs\Microsoft VS Code\bin\code" --wait
user.name=ChoongyeolKim
user.email=choongyeol.kim.79@gmail.com
(.venv) PS C:\Users\EL0022\Documents\20260324> 
```

A notification dialog box is open in the foreground, asking: "Would you like Visual Studio Code to periodically run 'git fetch'?" The dialog includes a "Source: Git" label and three buttons: "Yes", "No", and "Ask Me Later". The "Yes" button is highlighted with a red underline.

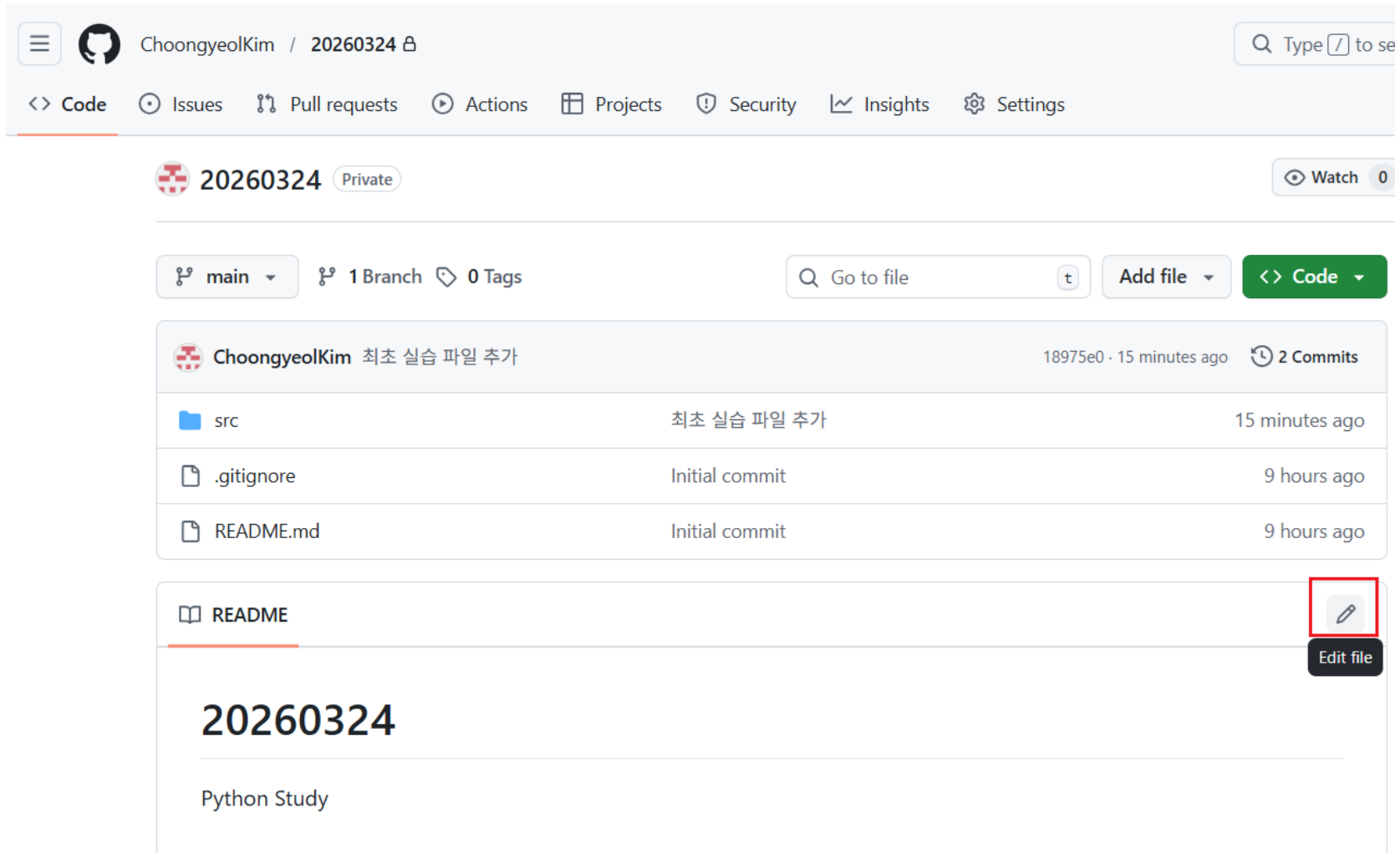
Git 원격 저장소에 push 3

commit 그래프가 변경되었음을 확인할 수 있고 웹에서도 확인해 봅니다.



Git 원격 저장소 pull 1

웹에서 README.md 파일을 수정합니다.



ChoongyeolKim / 20260324

Code Issues Pull requests Actions Projects Security Insights Settings

20260324 Private Watch 0

main 1 Branch 0 Tags Go to file Add file Code

ChoongyeolKim 최초 실습 파일 추가 18975e0 · 15 minutes ago 2 Commits

src	최초 실습 파일 추가	15 minutes ago
.gitignore	Initial commit	9 hours ago
README.md	Initial commit	9 hours ago

README

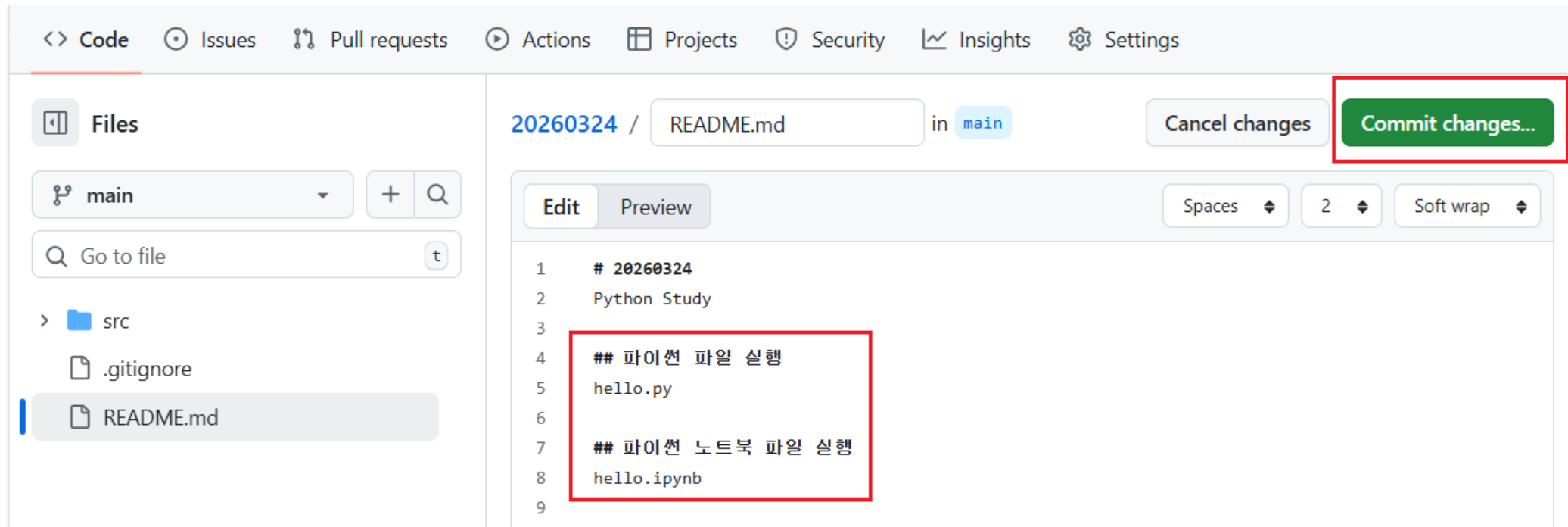
20260324

Python Study

Edit file

Git 원격 저장소 pull 2

내용을 수정한 후 commit 을 시도합니다.



Git 원격 저장소 pull 3

commit 메시지를 작성하고 commit 을 시도합니다.

Commit changes

Commit message

README.md 업데이트

Extended description

Add an optional extended description...

☒ Commit directly to the main branch

☐ Create a new branch for this commit and start a pull request [Learn more about pull requests](#)

Cancel

Commit changes

20260324 / README.md

ChoongyeolKim README.md 업데이트

Preview Code Blame

20260324

Python Study

파이썬 파일 실행

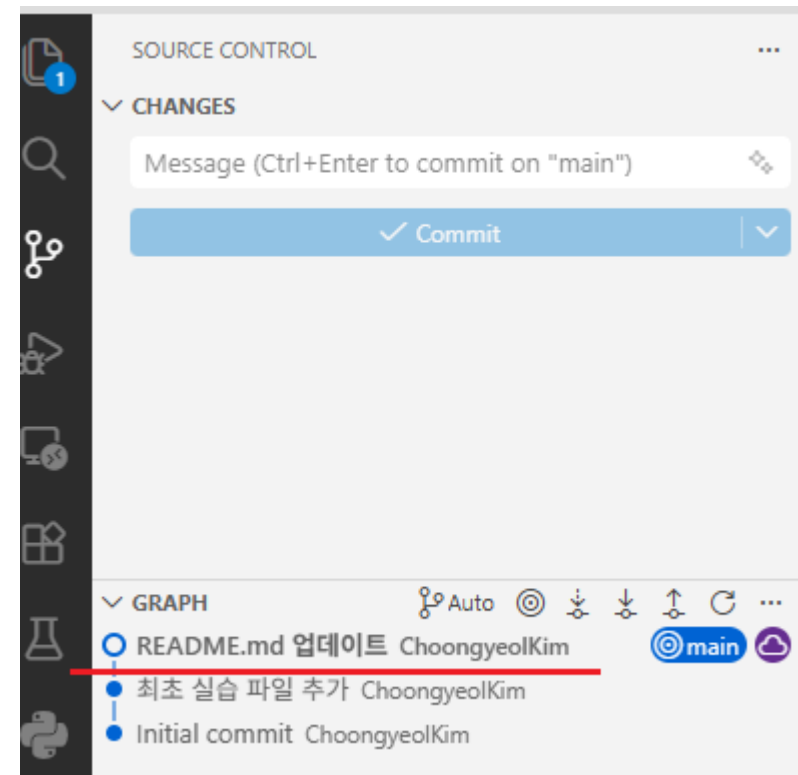
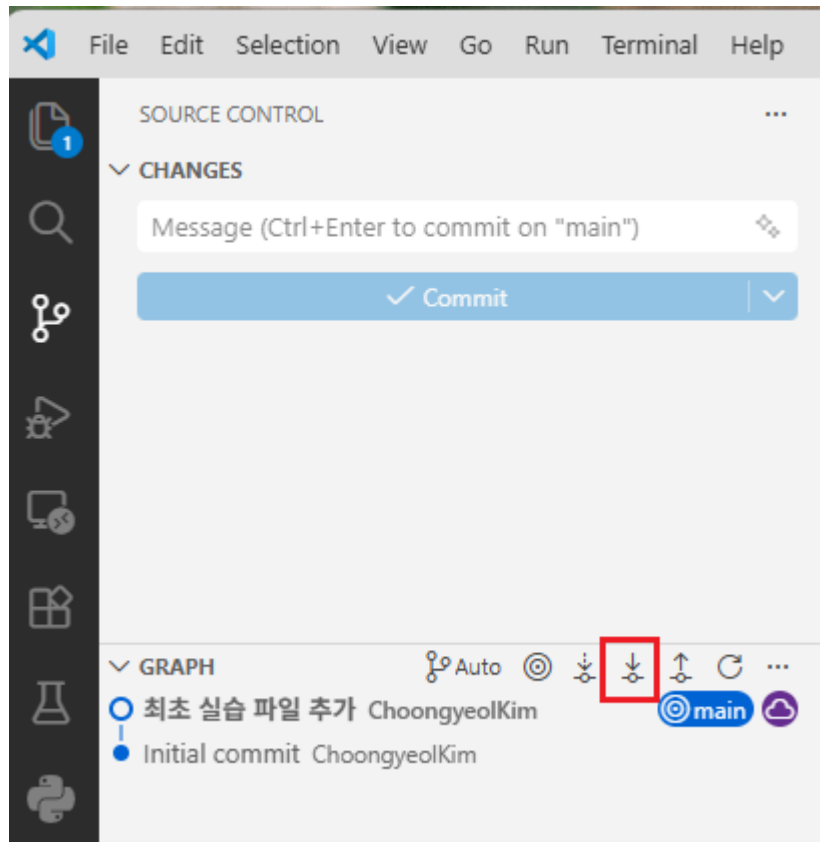
hello.py

파이썬 노트북 파일 실행

hello.ipynb

Git 원격 저장소 pull 4

원격 저장소에 변경사항이 발생했지만 로컬 저장소는 아직 변경사항이 반영되지 않았으므로 로컬 저장소를 pull 로 갱신해봅니다.

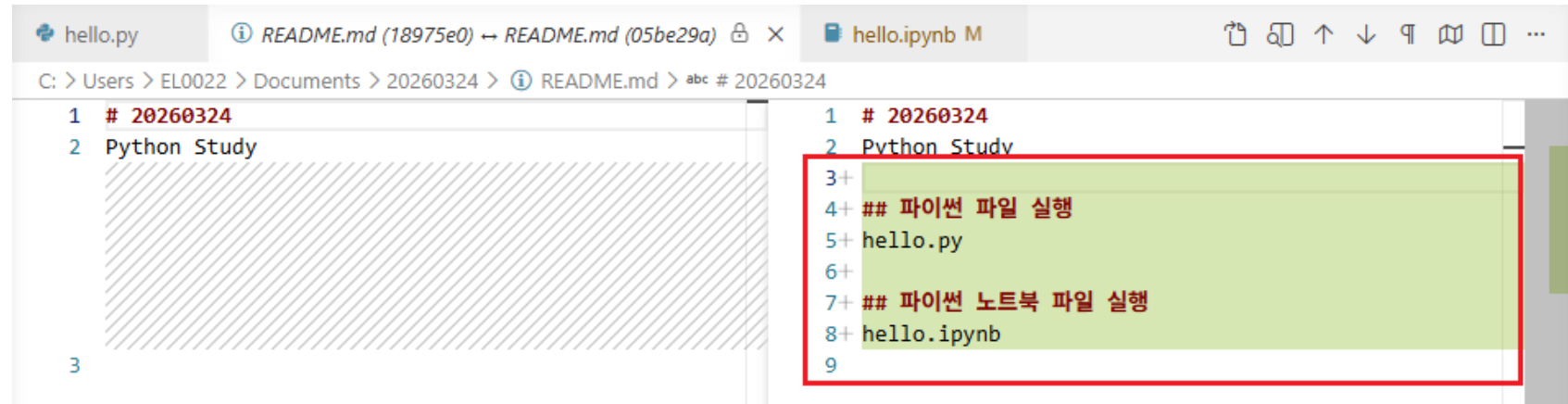
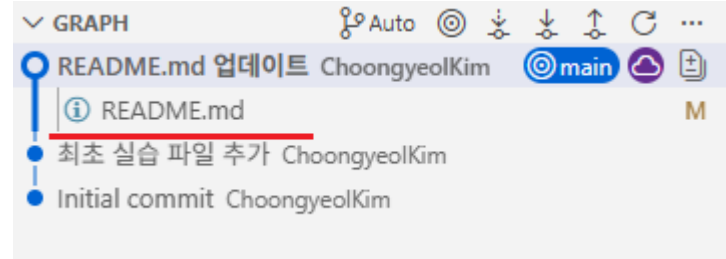
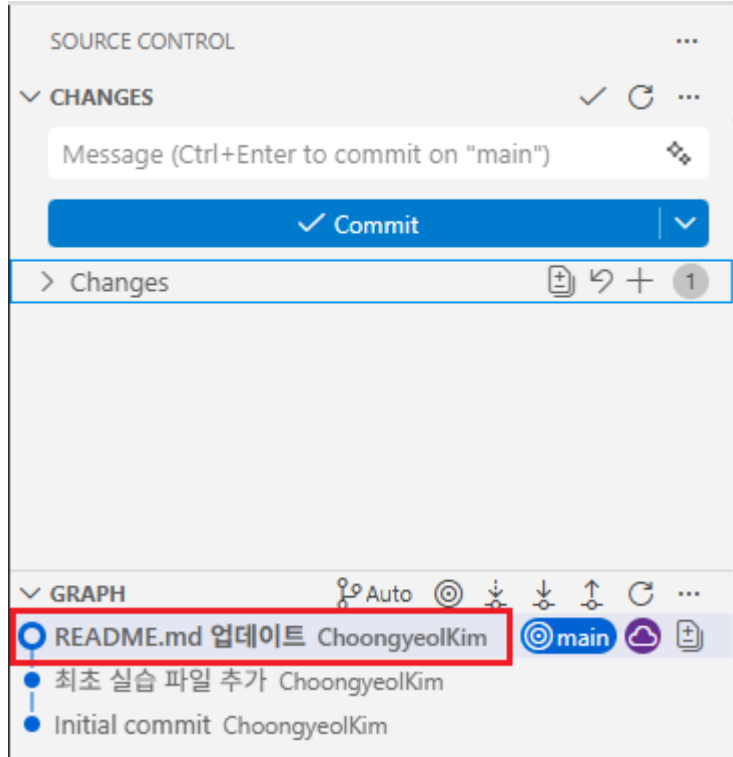


변경 사항 추적 (diff 확인)

마지막 커밋의 변경 사항을 확인해 봅니다.

마지막 커밋을 마우스로 클릭하면 해당 commit 에서 변경된 파일들이 전부 표시됩니다.

변경된 README.md 파일을 마우스로 클릭합니다.



더 알아야 할 것들

- Conflict (충돌)
 - Merge (병합)
 - Branch
 - Blame
 - Fetch 와 Pull 의 차이
 - Fast Forward (FF) 와 Rebase 의 차이
 - Pull Request (PR)
-
- 코드 리뷰
 - Git Branch 전략 – Git Flow, GitHub Flow
-
- 디버깅 (debugging)
 - 가상 환경 생성 명령어
 - 코딩 테스트

더 알아야 할 것들

- GUI Git 클라이언트 소프트웨어
- GUI Diff 소프트웨어
- 다른 코드 에디터 (Cursor, Antigravity etc)